Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

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Abstract

We propose a conformal prediction method for constructing tight simultaneous prediction intervals for multiple, potentially related, numerical outputs given a single input. This method can be combined with any multi-target regression model and guarantees finite-sample coverage. It is computationally efficient and yields informative prediction intervals even with limited data. The core idea is a novel *coordinate-wise* standardization procedure that makes residuals across output dimensions directly comparable, estimating suitable scaling parameters using the calibration data themselves. This does not require modeling of cross-output dependence nor auxiliary sample splitting. Implementing this idea requires overcoming technical challenges associated with transductive or full conformal prediction. Experiments on simulated and real data demonstrate this method can produce tighter prediction intervals than existing baselines while maintaining valid simultaneous coverage.

Keywords: interpretability, multivariate analysis, statistical inference, supervised learning, uncertainty quantification.

1 Introduction

1.1 Motivation and Preview of Contributions

We are interested in jointly quantifying uncertainty when predicting several, possibly dependent continuous outcomes from a shared set of input features. This problem is motivated by many applications where data-driven decisions depend on several unknown variables rather than on any one of them in isolation. For example, a financial institution may use available customer data to jointly forecast that customer's future credit-line utilization, monthly spending volume, and credit score. In this and many other contexts, decision quality hinges on the joint reliability of all predicted values, and an output that is incorrect in even one dimension could lead to adverse consequences.

Conformal prediction is a popular and broadly applicable framework for uncertainty quantification in predictive settings (Vovk et al., 2005; Angelopoulos and Bates, 2023). It derives its main strengths from the ability to augment the output of any machine-learning model with *prediction sets* equipped with finite-sample guarantees, without relying on strong assumptions on the data distribution. Its sole assumption is the availability of labeled data that are *exchangeable* with the test point of interest, a weaker requirement than the standard assumption of i.i.d. sampling. For a *univariate* setting, where the goal is to predict a single continuous-valued outcome, there now exist several well-established conformal prediction methods that produce valid, computationally fast, and easy-to-interpret prediction intervals (Lei et al., 2018; Romano et al., 2019; Sesia and Romano, 2021).

Extending conformal prediction to the *multivariate* setting is a non-trivial problem that has attracted considerable recent interest, spurring a variety of solutions (Dheur et al., 2025). The fundamental challenge is to aggregate a vector of generalized residuals—whose components may differ substantially in scale and variability—into a scalar-valued *non-conformity score* leading to *prediction sets* that should satisfy four key desiderata: (i) finite-sample coverage, (ii) informativeness or tightness, (iii) computational tractability, and (iv) interpretability. Interpretability can take different meanings depending on the application, and in some settings it is consistent with prediction sets having complex shapes. Nevertheless, in many practical scenarios, hyper-rectangular prediction sets—or, equivalently, jointly valid prediction intervals—provide the most natural and transparent way to convey uncertainty to practitioners. This paper focuses on such settings.

Existing methods achieve only a subset of these desiderata. The simplest approach to obtain hyper-rectangular prediction sets is to apply univariate conformal prediction to each output and then take the Cartesian product of the resulting intervals (Neeven and Smirnov, 2018); however, a conservative Bonferroni adjustment to the marginal miscoverage levels is needed to guarantee joint coverage; see Algorithm S1 in Appendix A. This strategy is very easy to implement, but the Bonferroni correction is often too conservative to be practically satisfactory, as it adopts a worst-case view of the dependence structure across output dimensions. By contrast, we aim to achieve valid coverage with joint prediction intervals that are as tight as possible; see Figure 1 for a sketch of this goal.

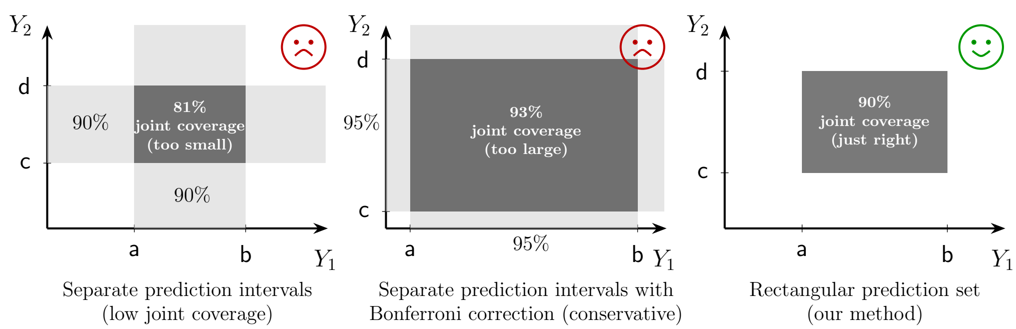

Figure 1: Two-dimensional toy example depicting graphically the main objective of this work: constructing *informative* and *interpretable* hyper-rectangular conformal prediction sets for multivariate outcomes. Left: univariate approaches can produce marginal prediction intervals for each output separately, which do not guarantee joint coverage. Center: a Bonferroni correction can easily restore valid joint coverage, at the cost of being overly conservative. Right: our goal is to efficiently construct hyper-rectangular prediction sets that are easy to interpret and retain valid coverage; doing so involves technical challenges.

A second class of existing methods reduce multivariate residuals to scalar-valued non-conformity scores, for example by applying an ℓ_2 or ℓ_∞ norm, then apply a univariate conformal prediction method to these scalar scores, and finally map the resulting prediction interval back into the original output space by inverting the dimension-reduction transformation; e.g., see Algorithm S2 in Appendix A. While computationally efficient and in principle able to produce hyper-rectangular prediction sets (e.g., using the ℓ_∞-norm), these approaches are dominated by the noisiest dimension, often leading to prediction sets that are unnecessarily wide along low-uncertainty directions.

More sophisticated approaches have been developed to directly minimize the volume of the resulting prediction sets; for example using optimal transport, learned embeddings, copula, or generative models to describe the dependence structure across residual dimensions. However, these methods aim to solve a problem that is different from that considered in this paper, since they are explicitly designed to produce non-rectangular prediction sets. Moreover, they often introduce substantial computational overhead and require additional independent training data to provide finite-sample guarantees.

In this paper we focus on targeting what we believe is the core difficulty in many applications: the mismatch in the *marginal* scale and variability of residuals across output dimensions. To address this challenge, we develop a method based on transductive coordinate-wise residual standardization. This restores comparability across coordinates and enables the construction of tight hyper-rectangular prediction sets that guarantee finite-sample coverage, without necessitating additional training data or prohibitive computational overhead. Remarkably, our method does *not* need to explicitly model residual dependencies across different dimensions, unlike existing approaches focusing on non-rectangular prediction sets.

Making this idea operational requires overcoming several hurdles, which we resolve through a sequence of increasingly practical steps. We begin by formulating an ideal oracle procedure that assumes access to the test point's true outcome vector and uses this information to standardize the multi-dimensional residuals in the calibration set without violating the exchangeability conditions that are necessary to guarantee finite-sample coverage. Although clearly impractical, this oracle is valuable to guide the design of our practical method. Inspired by the oracle procedure, we introduce two increasingly sophisticated data-driven approaches. The first provides a more conservative approximation of the oracle, whereas the second achieves greater statistical efficiency by building on top of the first. Through extensive numerical experiments, we demonstrate that this progression of ideas culminates in a practical method producing prediction sets satisfying all our desiderata.

1.2 Related Works

Extending conformal prediction to multivariate outcomes has attracted substantial attention in recent years, leading to the development of a wide range of alternative methods.

A prominent line of work seeks minimum-volume prediction sets with flexible geometries by modeling dependence across output dimensions. This includes copula-based approaches (Messoudi et al., 2021; Park and Cho, 2025; Sun and Yu, 2023), non-parametric density estimation methods (Sadinle et al., 2019; Izbicki et al., 2022; Wang et al., 2023; Dheur et al., 2025), parametric density estimation methods (Johnstone and Cox, 2021; Sampson and Chan, 2024; Braun et al., 2025a,b), optimal-transport techniques (Thurin et al., 2025; Klein et al., 2025), and flow-based generative models (Colombo, 2024; Lee et al., 2025; Fang et al., 2025). While these approaches can produce smaller prediction sets compared to our method, these often have complex shapes that are more difficult to interpret. Moreover, these approaches require additional model training on independent data to provide finite-sample guarantees, and many also involve substantial computational overhead.

To the best of our knowledge, the idea of directly addressing heterogeneity across output dimensions via marginal residual rescaling remains relatively under-explored. Baheri and Amiri Shahbazi (2025) pursue a related idea, but rely on a conservative Bonferroni-type correction. The approach of Sampson and Chan (2024) is more closely related to ours, but requires an additional independent dataset; avoiding this requirement is the main technical challenge addressed in our paper. As we show empirically, the method of Baheri and Amiri Shahbazi (2025) is often considerably less efficient than ours—particularly in small-sample regimes—resulting in less stable and less informative prediction sets.

Our solution can be viewed as implementing a special instance of transductive, or full, conformal prediction (Vovk et al., 2005; Vovk, 2013). In contrast to the more commonly used split conformal methods, full conformal prediction repeatedly incorporates the test point, augmented with different candidate labels, directly into a learning algorithm. Although this strategy is computationally prohibitive in general, especially for continuous-valued numerical outcomes and complex algorithms, it can be implemented efficiently in some cases by exploiting structural properties of the underlying predictive model. Accordingly, our paper relates to a broader literature on making full conformal inference computationally practical in special cases, including approaches tailored to sparse generalized linear models (Lei, 2019; Guha et al., 2023) and methods that leverage stability or other forms of structure in the predictive model (Ndiaye and Takeuchi, 2019; Abad et al., 2022; Martinez et al., 2023; Li, 2024). Unlike those works, which focus primarily on accelerating model refitting, we assume a fixed, pre-trained model and instead aim to improve efficiency by avoiding calibration data splitting while explicitly standardizing residuals across output dimensions. Both the problem setting and the proposed methodology are therefore novel.

More broadly, our approach is also related to recent conformal methods that reuse calibration data for different purposes, such as enabling early stopping to control overfitting when training deep learning models (Liang et al., 2023) and leveraging data-adaptive grouping to improve conditional coverage (Zhou and Sesia, 2024; Zhang and Candès, 2024). In contrast to these works, our focus is on addressing heterogeneity across targets in multivariate regression while obtaining interpretable, rectangular prediction sets.

1.3 Outline

This paper is organized as follows. Section 2 introduces notations, a formal problem statement, and necessary assumptions, along with a description of an ideal oracle approach. Section 3 presents our methodology, starting with a more conservative approach and then developing more efficient and computationally tractable implementations. Section 4 validates our method empirically and compares its performance to that of existing approaches using both simulated and real data. Section 5 discusses some limitations and possible directions for future research. The Appendices contain mathematical proofs not contained in the main text, additional algorithmic details, and further numerical results.

2 Problem Setup and Motivating Constructions

2.1 Notation

For any $n \in \mathbb{N}$, we denote $[n] := \{1, \ldots, n\}$. For any $m \in \mathbb{N}$, $\alpha \in (0, 1)$, and $(v_1, \ldots, v_m) \in \mathbb{R}^m$, let $\widehat{Q}_{1-\alpha}(v_1, \ldots, v_m)$ denote the $\lceil (1-\alpha)(m+1) \rceil$-th smallest element of $\{v_1, \ldots, v_m, +\infty\}$.

2.2 Setup and Data Assumptions

We consider a multi-output (multivariate) regression setting where the goal is to predict a vector $\mathbf{Y} = (Y_1, \ldots, Y_d)$ of d real-valued outputs given input features $\mathbf{X} \in \mathcal{X}$. Assume that $(\mathbf{X}^1, \mathbf{Y}^1), \ldots, (\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$ are jointly exchangeable, or drawn i.i.d. from an arbitrary distribution $P_{X,Y}$. We observe the calibration data $\mathcal{D}_{\text{cal}} = \{(\mathbf{X}^i, \mathbf{Y}^i)\}_{i=1}^n$ and the test covariate $\mathbf{X}^{n+1}$, while the corresponding outcome $\mathbf{Y}^{n+1}$ remains unobserved. Using all of the observed data, the task is to construct a reasonably tight prediction set $\hat{C}(\mathbf{X}^{n+1}) \subset \mathbb{R}^d$ that contains the unknown output $\mathbf{Y}^{n+1}$ with high probability, guaranteeing *marginal coverage*:

$$\mathbb{P}\Big[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})\Big] \geq 1 - \alpha, \tag{1}$$

where $\alpha \in (0, 1)$ is the user-specified miscoverage level. Note that both $\mathbf{Y}^{n+1}$ and $\hat{C}(\mathbf{X}^{n+1})$ are random in (1), as they depend on the new test point as well as on the calibration data $\mathcal{D}_{\text{cal}}$. The term "marginal coverage" reflects that the guarantee in (1) holds on average over the joint randomness of both calibration sample and the test point. Stronger notion of coverage, such as conditional coverage on which the probability in (1) is conditioned on some fixed values or ranges of the test point, are generally unattainable in finite samples without much more restrictive assumptions (Foygel Barber et al., 2021). There are some techniques to approximate the conditional coverage (Romano et al., 2020; Gibbs et al., 2025) that may be possible to combine with our approach and could be an interesting direction for future work, but it is beyond the scope of this paper.

For interpretability, we aim to enforce the prediction sets to be *hyper-rectangular*, or Cartesian product of univariate prediction intervals:

$$\hat{C}(\mathbf{X}^{n+1}) = \bigtimes_{j=1}^{d} \big[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})\big], \tag{2}$$

where $[L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})]$ denotes the coordinate-wise interval for the jth output. Prediction sets in this form decompose naturally across dimensions, making them very easy to explain in the context of multi-output prediction. Consequently, the marginal coverage requirement stated in (1) is equivalent to

$$\mathbb{P}\left[Y_j^{n+1} \in [L_j(\mathbf{X}^{n+1}), U_j(\mathbf{X}^{n+1})] \text{ for all } j \in [d]\right] \geq 1 - \alpha.$$

Thus, although the coverage guarantee is marginal over the calibration and test data, it is simultaneous across all d output coordinates.

2.3 Coordinate-Wise Generalized Residuals

In addition to the data assumptions, we also assume access to a *pre-trained* joint predictive model $\hat{\mathbf{f}} = (\hat{f}_1, \ldots, \hat{f}_d)$, fitted on dataset independent of both the calibration set $\mathcal{D}_{\text{cal}}$ and the test point. This model is treated as a "black-box" and could be anything; the only requirement for $\hat{\mathbf{f}}$ is that it must provide us with a coordinate-wise *generalized residual* map $\mathbf{E} : \mathcal{X} \times \mathbb{R}^d \to \mathbb{R}^d$, which quantifies the model's prediction error along each output dimension.

Let $\mathbf{E}(\mathbf{X}, \mathbf{Y}) = \big(E_1(\mathbf{X}, Y_1), \dots, E_d(\mathbf{X}, Y_d)\big) \in \mathbb{R}^d$ denote a vector of coordinate-wise generalized residuals. A classical example for models $\hat{f}_j$ that estimate the conditional mean $\mathbb{E}[Y_j \mid \mathbf{X}]$ (Lei et al., 2018) is $E_j(\mathbf{X}, Y_j) = |Y_j - \hat{f}_j(\mathbf{X})|$. Alternatively, for models that estimate conditional quantiles, as in conformalized quantile regression (Romano et al., 2019), a natural choice is

$$E_j(\mathbf{X}, Y_j) = \max\big\{\hat{f}_j^{\mathrm{lo}}(\mathbf{X}) - Y_j,\ Y_j - \hat{f}_j^{\mathrm{hi}}(\mathbf{X})\big\},$$

where $\hat{f}_j = (\hat{f}_j^{\mathrm{lo}}, \hat{f}_j^{\mathrm{hi}})$ denote the estimated lower and upper conditional quantiles of $Y_j \mid \mathbf{X}$, for example at levels $\alpha/2$ and $1 - \alpha/2$. Intuitively, this choice of E_j measures the signed distance of Y_j to the nearest boundary of the interval $[\hat{f}_j^{\mathrm{lo}}(\mathbf{X}), \hat{f}_j^{\mathrm{hi}}(\mathbf{X})]$, where positive values indicate that Y_j falls outside the interval, while negative values indicate that it lies inside.

Applying the generalized residual map to the observed calibration data gives $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i) = (E_1^i, \dots, E_d^i)$ for all $i \in [n]$. We reserve $\mathbf{E}^{n+1} = \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{Y}^{n+1})$ for the unobserved residual vector of the test point.

For the types of residuals with possibly negative coordinates, such as the quantile-based residuals reviewed above, there are at least two simple ways to adapt them to this assumption. If the residuals are bounded below by a fixed constant almost surely, non-negativity can be enforced by adding such constant to all calibration and test residuals—a transformation that can then be easily inverted to keep the resulting prediction sets invariant. If no such bound exists, negative residuals can instead be truncated at zero, at the cost of potentially more conservative conformal prediction sets (Romano et al., 2019).

We also impose two additional regularity conditions. First, each residual coordinate is assumed to have finite mean and finite, strictly positive variance at the population level. These moment conditions are mild and automatically satisfied by bounded and nondegenerate residuals. Second, within each coordinate, the calibration and test residuals are assumed to be almost surely distinct, which is a standard assumptionin conformal inference. When ties arise, including the ties at zero that may be introduced by truncation, they can be handled through standard randomized tie-breaking or by adding independent continuous jitter to the residuals. These conditions make the theoretical development cleaner, while our numerical experiments indicate that the proposed method remains robust when they are not exactly satisfied. We collect all of the standing conditions below.

Assumption 1 *For every $j \in [d]$, the distribution of the residual $E_j(\mathbf{X}, Y_j)$ satisfies $E_j(\mathbf{X}, Y_j) \geq 0$ almost surely, $\mathbb{E}[E_j(\mathbf{X}, Y_j)] < \infty$, and $0 < \mathrm{Var}[E_j(\mathbf{X}, Y_j)] < \infty$. Moreover, for every $j \in [d]$, the residuals $E_j^1, \dots, E_j^{n+1}$ are almost surely pairwise distinct:*

$$\mathbb{P}\left(E_j^i \neq E_j^k \text{ for all } 1 \leq i < k \leq n+1\right) = 1.$$

2.4 Conformal Calibration with Marginal Coverage

After evaluating the residuals $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i)$ for all calibration points indexed by $i \in [n]$, a prediction set for $\mathbf{Y}^{n+1}$ based on $\mathbf{X}^{n+1}$ can be assembled by including all possible values of

$\mathbf{y} \in \mathbb{R}^d$ whose residuals $\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})$ are sufficiently small:

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq W_j, \text{ for all } j \in [d] \right\}, \tag{3}$$

where $W_1, \ldots, W_d$ are suitable (data-dependent) thresholds chosen to satisfy

$$\mathbb{P}\left[0 \leq E_j(\mathbf{X}^{n+1}, Y_j^{n+1}) \leq W_j \text{ for all } j \in [d] \right] \geq 1 - \alpha. \tag{4}$$

This will ensure marginal coverage of $\mathbf{Y}^{n+1}$ at level $1-\alpha$, as defined in (1). For standard choice of $\mathbf{E}$, such as the absolute residuals, the resulting prediction set $\hat{C}(\mathbf{X}^{n+1})$ is hyper-rectangular, as in (2), with $L_j(\mathbf{X}^{n+1}) = \hat{f}_j(\mathbf{X}^{n+1}) - W_j$ and $U_j(\mathbf{X}^{n+1}) = \hat{f}_j(\mathbf{X}^{n+1}) + W_j$ for all $j \in [d]$. Similar for the quantile-based residuals described above.

A naive way to achieve (4) is to set $W_j = \hat{Q}_{1-\alpha/d}(E_j^1, \ldots, E_j^n)$, which is equivalent to applying univariate conformal prediction separately across all d outcome dimensions, with a Bonferroni correction (α/d replacing α). This typically leads to very conservative prediction sets because it takes an overly pessimistic worst-case view of possible dependencies of the prediction tasks across outcome dimensions.

In this paper, we pursue a more efficient approach, started by computing induced *scalar* non-conformity scores $S^i = \Phi(\mathbf{E}^i) \in \mathbb{R}$ for all $i \in [n]$ using a scalarization map $\Phi : \mathbb{R}_+^d \to \mathbb{R}$, whose form will be discussed below. Then, the prediction set is constructed as:

$$\hat{C}(\mathbf{X}^{n+1}) = \left\{ \mathbf{y} \in \mathbb{R}^d : \Phi(\mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y})) \leq \hat{Q}_{1-\alpha}(S^1, \ldots, S^n) \right\}. \tag{5}$$

If Φ is, in addition, an invertible map, then we can recover the form of (3) from (5) by setting $W_j = \Phi_j^{-1}\left(\hat{Q}_{1-\alpha}(S^1, \ldots, S^n) \right)$. We focus on the case where Φ is invertible in this paper, but the coverage guarantee is generally valid for other type of Φ as demonstrated in the following theorem.

Theorem 1 (Generalized Marginal Coverage) *Assume* $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ *are exchangeable, with* $\mathbf{E}^i = \mathbf{E}(\mathbf{X}^i, \mathbf{Y}^i)$ *for all* $i \in [n+1]$. *Let* $\Phi : \mathbb{R}_+^d \to \mathbb{R}$ *be any* fixed *function, independent of the data. Then,* $\hat{C}(\mathbf{X}^{n+1})$ *defined in* (5) *satisfies* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \geq 1-\alpha$. *Moreover, if the scores* $(S^1, \ldots, S^{n+1})$ *given by* $S^i = \Phi(\mathbf{E}^i)$ *are almost-surely distinct,* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \leq 1 - \alpha + \frac{1}{n+1}$. *The same results hold with* Φ *random, if* $(S^1, \ldots, S^{n+1})$ *are exchangeable conditional on it.*

If the induced scalar scores $S^1, \ldots, S^{n+1}$ are almost surely distinct, Theorem 1 guarantees the prediction sets defined in (5) achieve marginal coverage *tightly* from above, as long as the sample size n is not too small. A limitation of Theorem 1, however, is that it cannot tell us whether the coverage is equally tight across different outcome dimensions. In fact, depending on the data distribution, the fitted model, and the choice of Φ, it could happen that $\hat{C}(\mathbf{X}^{n+1})$ achieves the desired marginal coverage while being much wider than necessary along certain directions, thereby providing uncertainty estimates that are potentially much less informative than ideal. Moreover, depending on Φ, the prediction set $\hat{C}(\mathbf{X}^{n+1})$ may have a difficult-to-interpret shape.

Several recent works propose residual transformations based on ℓ_p-norms, i.e., $\Phi(\cdot) = \|\cdot\|_p$ for different $1 \leq p \leq \infty$ (Johnstone and Ndiaye, 2022). Although conceptually and computationally simple, these can be heavily affected by the noisiest output dimension. A single high-variance dimension may dominate the scalar score and inflate the resulting prediction set uniformly across all coordinates. To obtain tighter prediction sets, recent works have proposed learning a suitable transformation of the residuals prior to applying a norm-based dimension reduction, for example by using optimal transport (Thurin et al., 2025; Klein et al., 2025). However, these approaches require additional training data as well as expensive computations, and they typically lead to prediction sets with irregular shapes.

In this paper, we propose a conceptually simpler solution that consists of *standardizing* the residuals prior to reducing their dimension by taking the maximum across coordinates, similar to the ℓ_∞ norm. This leads to rectangular prediction sets that are uniformly tight across all output dimensions, without requiring additional training data or expensive computations. Despite the apparent simplicity of this idea, however, achieving all desiderata while maintaining guaranteed finite-sample coverage involves significant technical challenges, which we overcome as gradually explained below.

2.5 A Population-Standardized Oracle

To motivate our method, we first consider an oracle construction that has access to the coordinate-wise population means and standard deviations of the residuals. Although unavailable in practice, this construction illustrates how coordinate-wise standardization can mitigate heterogeneity in residual scales while preserving validity and rectangular interpretability.

Consider prediction sets of the form in (5), obtained by transforming the vector-valued residuals $\mathbf{E}^i$'s using $\Phi : \mathbb{R}_+^d \to \mathbb{R}$:

$$\Phi(\mathbf{t}; \boldsymbol{\beta}, \boldsymbol{\theta}) := \max_{1 \leq j \leq d} \frac{t_j - \beta_j}{\theta_j}, \tag{6}$$

where $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_d)^\top$ and $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_d)^\top$ are vectors of location and scale parameters, respectively. This transformation is invertible, with coordinate-wise inverse $\Phi_j^{-1}(c; \boldsymbol{\beta}, \boldsymbol{\theta}) = c \cdot \theta_j + \beta_j$ for any $c \in \mathbb{R}$. Given these two parameters, we define the $1-\alpha$ level prediction set of $\mathbf{X}^{n+1}$ by

$$\begin{aligned} C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta}) = \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha}^{\boldsymbol{\beta},\boldsymbol{\theta}} \cdot \theta_j + \beta_j \text{ for all } j \in [d] \right\}, \\ \text{where} \quad \hat{Q}_{1-\alpha}^{\boldsymbol{\beta},\boldsymbol{\theta}} := \hat{Q}_{1-\alpha}(S^1_{\boldsymbol{\beta},\boldsymbol{\theta}}, \ldots, S^n_{\boldsymbol{\beta},\boldsymbol{\theta}}) \quad \text{and} \quad S^i_{\boldsymbol{\beta},\boldsymbol{\theta}} = \Phi(\mathbf{E}^i; \boldsymbol{\beta}, \boldsymbol{\theta}). \end{aligned} \tag{7}$$

This set is hyper-rectangular and aligns the form of (3) with coordinate-wise thresholds $W_j = \Phi_j^{-1}(\hat{Q}_{1-\alpha}^{\boldsymbol{\beta},\boldsymbol{\theta}}; \boldsymbol{\beta}, \boldsymbol{\theta})$. Theorem 1 guarantees that $C(\mathbf{X}^{n+1}; \boldsymbol{\beta}, \boldsymbol{\theta})$ satisfies the marginal coverage requirement in (1) for any fixed choice of $(\boldsymbol{\beta}, \boldsymbol{\theta})$.

To address the possible heterogeneity in scale and variability across different dimensions, the most natural option of $(\boldsymbol{\beta}, \boldsymbol{\theta})$ are the population mean and standard deviation of the residual vector $\mathbf{E}(\mathbf{X}, \mathbf{Y})$, which we denote by $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$:

$$\boldsymbol{\mu}_j^* = \mathbb{E}[E_j(\mathbf{X}, Y_j)] \geq 0, \qquad \boldsymbol{\sigma}_j^* = \sqrt{\mathrm{Var}[E_j(\mathbf{X}, Y_j)]} > 0, \qquad \forall j \in [d]. \tag{8}$$

Then by replacing $(\boldsymbol{\beta}, \boldsymbol{\theta})$ in (7) with the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$, we obtain a population-standardized oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$.

Although practically unfeasible, this oracle prediction set $C(\mathbf{X}^{n+1}; \boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ satisfies all four of our desiderata: (i) it guarantees finite-sample marginal coverage, as per Theorem 1; (ii) it tends to be uniformly tight across all output dimensions, because it operates with residuals whose components are all on the same scale; (iii) it is fast to compute, at least for standard choices of the residual function $\mathbf{E}$; (iv) it is an easy-to-interpret hyper-rectangle, at least for standard choices of $\mathbf{E}$.

In the following, we will describe practical approximations of these oracle prediction sets, starting from intuitive but not fully satisfactory solutions before presenting our method.

2.6 Simple but Unsatisfactory Practical Approaches

An intuitive *plug-in* approximation of the ideal oracle is obtained by replacing the population parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ with their empirical counterparts $(\hat{\boldsymbol{\mu}}^{\mathrm{pi}}, \hat{\boldsymbol{\sigma}}^{\mathrm{pi}})$, evaluated on the observed calibration data, and then plugging in $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\mathrm{pi}}, \hat{\boldsymbol{\sigma}}^{\mathrm{pi}})$ into the prediction set formula (7): $\hat{C}^{\mathrm{pi}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\mathrm{pi}}, \hat{\boldsymbol{\sigma}}^{\mathrm{pi}})$. This approach, outlined by Algorithm S3 in Appendix A, tends to perform well in large-samples (as shown empirically in Appendix C) but lacks finite-sample guarantees. In fact, using a data-dependent transformation Φ, which breaks the exchangeability between calibration and test scores, violates the key assumption of Theorem 1.

An alternative solution that is also intuitive yet unsatisfactory consists of combining the above plug-in approach with an additional data splitting step that allows one to recover finite-sample guarantees: the location and scale parameter estimates for the transformation function Φ, denoted here as $\hat{\boldsymbol{\mu}}^{\mathbf{ds}}$ and $\hat{\boldsymbol{\sigma}}^{\mathbf{ds}}$, are estimated empirically using only a random subset of $n_1 < n$ observations of the observed calibration data. The remaining $n_2 = n - n_1$ data points are used to evaluate the quantile $\hat{Q}_{1-\alpha}^{\hat{\boldsymbol{\mu}}^{\mathrm{ds}}, \hat{\boldsymbol{\sigma}}^{\mathrm{ds}}}$ and construct $\hat{C}^{\mathrm{ds}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\mathrm{ds}}, \hat{\boldsymbol{\sigma}}^{\mathrm{ds}})$. This approach is outlined by Algorithm S4 in Appendix A. While it satisfies the key assumption of Theorem 1, it makes an inefficient use of the data. Since it reduces the effective number of calibration samples roughly by half, it tends to produce more unstable and often wider-than-necessary prediction sets, especially in small-sample regimes.

The main contribution of this paper is therefore to overcome this inefficiency. We develop a *transductive* conformal prediction method that uses a similar standardization idea but avoids wasteful data splitting by reusing the entire calibration data to estimate the scaling parameters, all while retaining finite-sample guarantees and manageable computational costs. Achieving this requires resolving several technical challenges, as explained step by step below.

3 Methodology

3.1 A Transductive Rescaling Oracle and High-Level Method Blueprint

According to Theorem 1, prediction sets of the form (5) achieve finite-sample marginal coverage if the dimension-reduction map Φ is independent of the calibration data, as assumed in the previous section, or if $(\Phi(\mathbf{E}^1), \ldots, \Phi(\mathbf{E}^{n+1}))$ can remain *exchangeable conditional on* Φ.

This latter requirement, which is strictly weaker than independence, is the foundation of the method developed in this section. We begin by showing how this conditional-exchangeability condition naturally leads to a different *transductive* oracle, which is much more closely aligned with the practical method introduced below.

Consider an imaginary oracle that knows the unordered values in $\{\mathbf{E}^1, \dots, \mathbf{E}^{n+1}\}$. We can estimate the population location and scale parameters $(\boldsymbol{\mu}^*, \boldsymbol{\sigma}^*)$ defined in (8) as follows:

$$\begin{aligned} \hat{\boldsymbol{\mu}}^{\text{oracle}} &= (\hat{\mu}_1^{\text{oracle}}, \dots, \hat{\mu}_d^{\text{oracle}})^\top, \quad \hat{\boldsymbol{\sigma}}^{\text{oracle}} = (\hat{\sigma}_1^{\text{oracle}}, \dots, \hat{\sigma}_d^{\text{oracle}})^\top, \\ \hat{\mu}_j^{\text{oracle}} &= \frac{1}{n+1} \sum_{i=1}^{n+1} E_j^i, \quad \hat{\sigma}_j^{\text{oracle}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n+1} \left(E_j^i - \hat{\mu}_j^{\text{oracle}}\right)^2}. \end{aligned} \tag{9}$$

Since these statistics are *symmetric* in all $n+1$ data points and thus invariant to their ordering, conditioning on the random function $\Phi^{\text{oracle}}(t) := \Phi(t; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ maintains the exchangeability of $(S_{\text{oracle}}^1, \dots, S_{\text{oracle}}^{n+1})$ where $S_{\text{oracle}}^i := \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$. Therefore, Theorem 1 implies that the prediction set $C^{\text{oracle}}(\mathbf{X}^{n+1}) := C(\mathbf{X}^{n+1}; \hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$, obtained by substituting $(\boldsymbol{\beta}, \boldsymbol{\theta}) = (\hat{\boldsymbol{\mu}}^{\text{oracle}}, \hat{\boldsymbol{\sigma}}^{\text{oracle}})$ into (7), satisfies the marginal coverage requirement in (1). Of course, this oracle is still impractical because both Φ^{oracle} and $\hat{Q}_{1-\alpha}^{\text{oracle}} := \hat{Q}_{1-\alpha}(S_{\text{oracle}}^1, \dots, S_{\text{oracle}}^n)$ depend on the unknown test outcome $\mathbf{Y}^{n+1}$ through $\mathbf{E}^{n+1}$.

We will now translate this oracle into a practical method producing a (slightly) larger prediction set that depends only on the observed data. First, we will find suitable *monotonically nondecreasing link functions* $\omega_1, \dots, \omega_d : \mathbb{R} \to \mathbb{R}_+$ satisfying

$$S_{\text{oracle}}^{n+1} \le c \iff 0 \le E_j^{n+1} \le \omega_j(c) \text{ for all } j \in [d], \tag{10}$$

for every $c \in \mathbb{R}$. Using these link functions, we can define an estimated $1-\alpha$ level prediction set of $\mathbf{X}^{n+1}$ by

$$\hat{C}(\mathbf{X}^{n+1}; c) = \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \omega_j(c) \text{ for all } j \in [d] \right\}. \tag{11}$$

Substituting $c = \hat{Q}_{1-\alpha}^{\text{oracle}}$ into (10) and (11) gives the equivalence relation

$$\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1}) \iff \mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}}). \tag{12}$$

Although $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ is still impractical, it is closer to being computable. The last remaining step is to replace $\hat{Q}_{1-\alpha}^{\text{oracle}}$ with a conservative data-driven estimate, which we denote here as $\hat{Q}_{1-\alpha}^{\text{prac}} \ge \hat{Q}_{1-\alpha}^{\text{oracle}}$. Because $\omega_1(c), \dots, \omega_d(c)$ are non-decreasing in c, doing that will give a practical prediction set

$$\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{prac}}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}}), \tag{13}$$

hence inheriting the coverage lower bound from Theorem 1. Technical difficulties, which we address below, arise when one tries to keep this approximation tight and computationally efficient.

3.2 Explicit Form of the Link Functions

The link functions $\omega_1, \ldots, \omega_d$ defined implicitly in (10) can be derived by solving the inequalities in (10) for E_j^{n+1} in terms of c, through the observed statistics $\hat{\mu}_j^{\text{obs}}$ and $\hat{\sigma}_j^{\text{obs}}$ defined as

$$\hat{\mu}_j^{\text{obs}} = \frac{1}{n}\sum_{i=1}^{n} E_j^i, \quad \hat{\sigma}_j^{\text{obs}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(E_j^i - \hat{\mu}_j^{\text{obs}}\right)^2}, \quad \forall j \in [d]. \tag{14}$$

The explicit form of $\omega_j(c)$ is given in the following lemma.

Lemma 2 *Under Assumption 1, suitable link functions $\omega_1, \ldots, \omega_d$ satisfying* (10) *are:*

$$\omega_j(c) = \begin{cases} 0, & \text{if } c \leq -\frac{n}{\sqrt{n+1}}, \\ \max\left\{0,\ \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\dfrac{(n+1)^2}{n^2-(n+1)c^2}}\right\}, & \text{if } -\dfrac{n}{\sqrt{n+1}} < c < 0, \\ \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}}|c|\sqrt{\dfrac{(n+1)^2}{n^2-(n+1)c^2}}, & \text{if } 0 \leq c < \dfrac{n}{\sqrt{n+1}}, \\ \infty, & \text{if } c \geq \dfrac{n}{\sqrt{n+1}}, \end{cases} \quad \forall j \in [d].$$

Replacing this expression for the link functions in (11) and substituting $c = \hat{Q}_{1-\alpha}^{\text{oracle}}$ give an upper bound $\omega_j(\hat{Q}_{1-\alpha}^{\text{oracle}})$ for $E_j(\mathbf{X}^{n+1}, y_j)$ that is finite (and hence informative) if and only if $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$. Fortunately, $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely as long as the sample size is not too small.

Lemma 3 *Under Assumption 1, if $n \geq 1/\alpha - 1$, then $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost surely.*

Armed with Lemma 2 and Lemma 3, what remains to be done is to find a tight upper bound $\hat{Q}_{1-\alpha}^{\text{prac}}$ for $\hat{Q}_{1-\alpha}^{\text{oracle}}$, as previewed in (13). This is the main challenge and focus of the next sections.

3.3 A Conservative Global-Worst-Case Approach

We first introduce a (highly) conservative practical upper bound $\hat{Q}_{1-\alpha}^{\text{gwc}}$, obtained by taking the worst-case upper bound of $\hat{Q}_{1-\alpha}^{\text{oracle}}$ over all possible values of $\mathbf{E}^{n+1} \in \mathbb{R}_+^d$. We refer to the resulting *global-worst-case* (GWC) prediction set as $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{gwc}})$. As we show in the next section, this conservative prediction set can in fact be sharpened considerably, without sacrificing marginal coverage, albeit with some additional methodological effort.

Let $\mathbf{z} \in \mathbb{R}_+^d$ denote a placeholder for the test residual $\mathbf{E}^{n+1}$. Define the plug-in scaling parameters $\hat{\boldsymbol{\mu}}(\mathbf{z}) := (\hat{\mu}_1(z_1), \ldots, \hat{\mu}_d(z_d))$ and $\hat{\boldsymbol{\sigma}}(\mathbf{z}) := (\hat{\sigma}_1(z_1), \ldots, \hat{\sigma}_d(z_d))$ as if $\mathbf{E}^{n+1}$ in (9) were replaced by $\mathbf{z}$:

$$\hat{\mu}_j(z_j) = \frac{\sum_{i=1}^{n} E_j^i + z_j}{n+1}, \quad \hat{\sigma}_j(z_j) = \sqrt{\frac{\sum_{i=1}^{n}\left(E_j^i - \hat{\mu}_j(z_j)\right)^2 + \left(z_j - \hat{\mu}_j(z_j)\right)^2}{n}}. \tag{15}$$

With this notation, we have $\hat{\boldsymbol{\mu}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\mu}}^{\text{oracle}}$ and $\hat{\boldsymbol{\sigma}}(\mathbf{E}^{n+1}) = \hat{\boldsymbol{\sigma}}^{\text{oracle}}$. Define then the GWC transformation $\hat{\Phi}^{\text{gwc}} : \mathbb{R}_+^d \to \mathbb{R}$ as

$$\hat{\Phi}^{\text{gwc}}(\mathbf{t}) := \max_{1 \leq j \leq d} \sup_{z_j \geq 0} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \geq \Phi^{\text{oracle}}(\mathbf{t}), \qquad \forall \mathbf{t} \in \mathbb{R}_+^d. \tag{16}$$

Intuitively, for each coordinate $j \in [d]$ and $\mathbf{t} \in \mathbb{R}_+^d$, we consider the most unfavorable (largest) possible value of the standardized ratio as a function of $E_j^{n+1} = z_j \geq 0$. This ensures that $\hat{\Phi}^{\text{gwc}} \geq \Phi^{\text{oracle}}$ uniformly. Conveniently, $\hat{\Phi}^{\text{gwc}}$ admits a closed-form expression.

Lemma 4 *The GWC transformation $\hat{\Phi}^{\text{gwc}}$ defined in* (16) *can be written as:*

$$\hat{\Phi}^{\text{gwc}}(\mathbf{t}) = \max_{1 \leq j \leq d} \max \left\{ \frac{t_j - \hat{\mu}_j(0)}{\hat{\sigma}_j(0)}, \frac{t_j - \hat{\mu}_j(z_j^*)}{\hat{\sigma}_j(z_j^*)}, -\frac{1}{\sqrt{n+1}} \right\}.$$

where $z_j^* = \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}$ if $\hat{\mu}_j^{\text{obs}} \geq \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}$ and $z_j^* = 0$ otherwise.

Using Lemma 4, we compute the scalar non-conformity scores $S_{\text{gwc}}^i := \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ for $i \in [n]$, and define the corresponding conformal quantile $\hat{Q}_{1-\alpha}^{\text{gwc}} := \hat{Q}_{1-\alpha}(S_{\text{gwc}}^1, \ldots, S_{\text{gwc}}^n)$. Because $\hat{\Phi}^{\text{gwc}} \geq \Phi^{\text{oracle}}$, it follows that $\hat{Q}_{1-\alpha}^{\text{gwc}} \geq \hat{Q}_{1-\alpha}^{\text{oracle}}$ almost surely. Substituting $c = \hat{Q}_{1-\alpha}^{\text{gwc}}$ in (11), we obtain a practical prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{gwc}})$ that is guaranteed to cover the unknown test outcome $\mathbf{Y}^{n+1}$ with probability at least $1 - \alpha$. Algorithm S5 in Appendix A summarizes this procedure.

Theorem 5 *Assume $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ are exchangeable and Assumption 1 holds. Then, $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ constructed by Algorithm S5 satisfies: $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})] \geq 1 - \alpha$.*

Proof From (11), it is easy to see that $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ almost surely because $\hat{Q}_{1-\alpha}^{\text{gwc}} \geq \hat{Q}_{1-\alpha}^{\text{oracle}}$ and the link functions $\omega_1, \ldots, \omega_d$ outlined in Lemma 2 are monotonically non-decreasing. Then Theorem 1 and the dual relation in (12) together imply that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})] = \mathbb{P}[\mathbf{Y}^{n+1} \in C^{\text{oracle}}(\mathbf{X}^{n+1})] \geq 1 - \alpha$. ■

We refer to Appendix A.3.1 for details on how to construct $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ in practice at cost $\mathcal{O}(dn)$. While computationally efficient, however, $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ is *statistically* inefficient as it tends to be substantially larger than the estimated oracle prediction set $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$, which we would ideally like to approximate tightly. In the next section, we therefore extend this approach and develop a tighter *local worst-case* (LWC) prediction set $\hat{C}_{\mathcal{P}}^{\text{lwc}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ that approximates the oracle $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ more closely.

3.4 A Tighter Local-Worst-Case Method

The construction of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ described in the previous section relies on a *global* upper bound $\hat{Q}_{1-\alpha}^{\text{gwc}}$ for the ideal quantile $\hat{Q}_{1-\alpha}^{\text{oracle}}$, designed to hold uniformly over all possible values of the unobserved test residual $\mathbf{E}^{n+1} \in \mathbb{R}_+^d$. However, the resulting prediction sets are often overly conservative, as the GWC procedure treats all realizations of $\mathbf{E}^{n+1}$ equally, even if some choices correspond to $\mathbf{y}$'s that are extremely unlikely to satisfy the condition stated in $\hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha})$ and could therefore be safely discarded.

This motivates the development of a more statistically efficient approach that utilizes a collection of *local* prediction sets, each tailored to a distinct working hypothesis that the true outcome $\mathbf{Y}^{n+1}$ belongs to a specific region of the outcome space. We will first present the high-level overview of this method and then detail the implementation in three steps: (1) we introduce an intuitive, data-driven partition of the outcome space guided by the order statistics of the calibration residuals; (2) we derive a tractable explicit characterization of the local prediction sets; (3) we develop a computationally efficient algorithm for constructing a tight enclosing rectangular prediction set in practice, while avoiding explicit enumeration of an exponential number of local prediction sets.

3.4.1 Method Overview

For any subset $\mathcal{R} \subseteq \mathbb{R}^d$, we consider the working hypothesis that $\mathbf{Y}^{n+1} \in \mathcal{R}$. Let $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})$ denote an upper bound for $\hat{Q}^{\text{oracle}}_{1-\alpha}$ tailored to this hypothesis, which satisfies

$$\mathbf{Y}^{n+1} \in \mathcal{R} \quad \Longrightarrow \quad \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R}) \geq \hat{Q}^{\text{oracle}}_{1-\alpha}. \tag{17}$$

Substituting $c = \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})$ in (11), we get a local prediction set

$$\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{R})) \cap \mathcal{R} \tag{18}$$

that satisfies

$$\mathbf{Y}^{n+1} \in \mathcal{R} \implies \mathcal{R} \supseteq \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \supseteq \hat{C}(\mathbf{X}^{n+1}; \hat{Q}^{\text{oracle}}_{1-\alpha}) \cap \mathcal{R}. \tag{19}$$

It is worth noting that the containment chain on the right is generally not guaranteed if $\mathbf{Y}^{n+1} \notin \mathcal{R}$. Then Theorem 1 and the dual relation (12) implies that $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ is guaranteed to cover the unknown test outcome $\mathbf{Y}^{n+1}$ with probability at least $1-\alpha$, conditional on the working hypothesis $\mathbf{Y}^{n+1} \in \mathcal{R}$.

Of course, we do not have the extra information on which fixed region $\mathcal{R}$ contains $\mathbf{Y}^{n+1}$ in practice, but if we can find a sufficiently rich collection of regions $\mathcal{P}$ such that $\mathbf{Y}^{n+1}$ belongs to one of these regions with high probability, we can obtain a valid prediction set by aggregating the local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ over all $\mathcal{R} \in \mathcal{P}$. Furthermore, to ensure interpretability, we report a relatively tight rectangular set that bounds the union of all localized prediction sets, rather than the union itself.

Thanks to our previous GWC construction, we already have a valid candidate for such high-probability regions, namely the global-worst-case prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Let $\mathcal{P}$ be a *random partition* of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. In summary, we construct $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ so that it encloses the union of all local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ for $\mathcal{R} \in \mathcal{P}$ with a rectangle, while remaining contained in the global-worst-case prediction set $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Letting $\text{Rect}_{\mathcal{P}}(A)$ denote a rectangle that contains the set A while being contained in $\bigsqcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R} = \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$, we define the local-worst-case prediction set as

$$\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) := \text{Rect}_{\mathcal{P}}\left(\bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \right) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}). \tag{20}$$

In practice, our method constructs a *relatively tight*, though not necessarily the *tightest*, such rectangle due to computational considerations. Nonetheless, the resulting prediction sets are (often much) more informative than $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ in finite samples. Moreover, in the large-sample limit, the union in (20) becomes approximately rectangular itself, making this enclosing step a very mild relaxation from the perspective of statistical efficiency.

The following result guarantees that this high-level approach leads to prediction set $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ that retains valid marginal coverage.

Proposition 6 *Consider a (random) collection $\mathcal{P}$ of disjoint regions $\mathcal{R} \subseteq \mathbb{R}^d$ satisfying $\sqcup_{\mathcal{R}\in\mathcal{P}}\mathcal{R} = \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ almost surely. For any $\mathcal{R} \in \mathcal{P}$, let $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1};\mathcal{R}) \subseteq \mathbb{R}^d$ denote a local prediction set satisfying* (19) *almost surely. Then, $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ defined in* (20) *satisfies $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$. Moreover, if $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ are exchangeable and Assumption 1 holds, $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})] \geq 1-\alpha$.*

We next introduce an intuitive data-driven partition based on the possible rankings of the test residuals E_j^{n+1} relative to the calibration residuals $(E_j^1, \ldots, E_j^n)$ for each coordinate $j \in [d]$, and then we characterize the corresponding prediction set $\hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$.

3.4.2 Step 1: A Data-Driven Partition Based on Sample Order Statistics

For each $j \in [d]$, let $E_j^{(k)}$ denote the k-th order statistic of $\{E_j^i\}_{i=1}^n$, so that $0 := E_j^{(0)} < E_j^{(1)} \leq \ldots \leq E_j^{(n)} < E_j^{(n+1)} := +\infty$. For each multi-index $h = (h_1, \ldots, h_d) \in [n+1]^d$, define the local region $\mathcal{Y}^h$ in the outcome space as:

$$\begin{aligned} &\mathcal{Y}^h := \left\{ \mathbf{y} \in \mathbb{R}^d : \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{E}^h \right\}, \quad \text{where} \\ &\mathcal{E}^h := \bigtimes_{j=1}^d \big[\, E_j^{(h_j-1)}, U_j^{h_j} \,\big) \quad \text{and} \quad U_j^{h_j} := \min\left\{ E_j^{(h_j)}, \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha}) \right\}. \end{aligned} \tag{21}$$

Clearly $\mathcal{P}_{n,d} = \{\mathcal{Y}^h\}_{h\in[n+1]^d}$ partitions $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$ into $(n+1)^d$ disjoint regions, some of which may be empty depending on $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$. Figure 2 visualizes this partition and the corresponding construction of $\hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$, in a two-dimensional toy example.

While it may appear at first sight that this choice of partition necessarily incurs exponential computational costs of order $\mathcal{O}(n^d)$, rendering it intractable for all but very small dimensions d, in fact it enables a much more efficient characterization of the local prediction sets $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1};\mathcal{Y}^h)$ that will lead to a practical algorithm for constructing $\hat{C}^{\text{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})$ at a computational cost of only $\mathcal{O}(d^2 n \log n)$.

3.4.3 Step 2: Characterization of the Local Prediction Sets

For any $h \in [n+1]^d$ with $\mathcal{Y}^h \neq \emptyset$, we characterize the local prediction set $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1};\mathcal{Y}^h)$ by computing the scalar quantity $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h)$ that satisfies the constraint (17), following a strategy similar to the worst-case approach from Section 3.3.

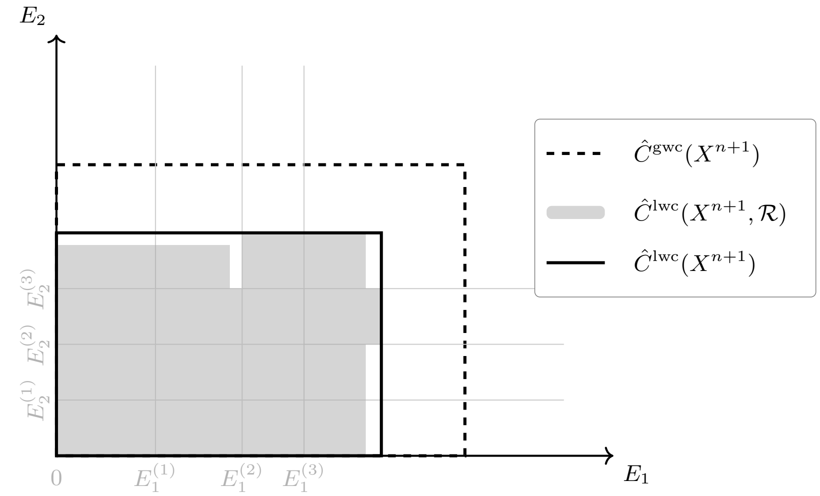

Figure 2: Illustration of the construction of transductively standardized conformal prediction sets using the method outlined in (20), with the data-driven partition $\mathcal{P}_{n,d}$ of the outcome space defined in (21), in a toy example with a two-dimensional outcome variable. The gray grid lines indicate the partition $\mathcal{P}_{n,d}$ in the residual space, as determined by the order statistics $E_j^{(k)}$ of the calibration residuals. The shaded rectangles represent local prediction sets $\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ corresponding to different hypothesized rectangles $\mathcal{R} \in \mathcal{P}_{n,d}$. The solid rectangle shows the projection, in residual space, of their rectangular enclosure defined in (20), which constitutes our final prediction set. For comparison, the larger dashed rectangle shows the corresponding projection of the more conservative prediction set $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ from Section 3.3.

Define the LWC scalarization map $\hat{\Phi}_h^{\mathrm{lwc}} : \mathbb{R}_+^d \to \mathbb{R}$ such that

$$\hat{\Phi}_h^{\mathrm{lwc}}(\mathbf{t}) := \max_{1 \le j \le d} \left\{ \sup_{\mathbf{z} \in \mathcal{E}^h} \frac{t_j}{\hat{\sigma}_j(z_j)} - \inf_{\mathbf{z} \in \mathcal{E}^{\mathrm{gwc}}} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \right\} \tag{22}$$

where $\mathcal{E}^{\mathrm{gwc}} := \bigtimes_{j=1}^d \left[0,\, \omega_j(\hat{Q}_{1-\alpha}^{\mathrm{gwc}})\right]$ is the projection of $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ onto the residual space that satisfies $\mathbf{y} \in \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1}) \iff \mathbf{E}(\mathbf{X}^{n+1}, \mathbf{y}) \in \mathcal{E}^{\mathrm{gwc}}$. With this choice, we have

$$\mathbf{E}^{n+1} \in \mathcal{E}^h \implies \hat{\Phi}_h^{\mathrm{lwc}}(\mathbf{t}) \ \ge\ \max_{1 \le j \le d} \sup_{\mathbf{z} \in \mathcal{E}^h} \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \ge \Phi^{\mathrm{oracle}}(\mathbf{t}), \quad \text{for all } \mathbf{t} \in \mathbb{R}_+^d. \tag{23}$$

By (21), this ensures that for any $h \in [n+1]^d$ with $\mathbf{Y}^{n+1} \in \mathcal{Y}^h \neq \emptyset$, $\hat{\Phi}_h^{\mathrm{lwc}} \ge \Phi^{\mathrm{oracle}}$ uniformly. Moreover, $\hat{\Phi}_h^{\mathrm{lwc}}(\mathbf{t})$ admits the following closed form.

Lemma 7 *If $\mathcal{Y}^h \neq \emptyset$, then $\hat{\Phi}_h^{\mathrm{lwc}}(\mathbf{t})$ defined in (22) can be written for any $\mathbf{t} \in \mathbb{R}_+^d$ as:*

$$\hat{\Phi}_h^{\mathrm{lwc}}(\mathbf{t}) = \max_{1 \le j \le d} \left\{ \frac{t_j}{r_j(h_j)} - \min\left\{ \frac{\hat{\mu}_j(0)}{\hat{\sigma}_j(0)},\ \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\mathrm{gwc}})} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \right\} \right\},$$

where $r_j(h_j) = \hat{\sigma}_j^{\mathrm{obs}}$ if $E_j^{(h_j-1)} \le \hat{\mu}_j^{\mathrm{obs}} < U_j^{h_j}$, and otherwise $\min\left\{\hat{\sigma}_j(E_j^{(h_j-1)}), \hat{\sigma}_j(U_j^{h_j})\right\}$.

Note that Lemma 7 involves a limit for $z_j \uparrow \omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})$ because $\omega_j(\hat{Q}^{\text{gwc}}_{1-\alpha})$ can be infinite when $\hat{Q}^{\text{gwc}}_{1-\alpha} \geq \frac{n}{\sqrt{n+1}}$ (recall Lemma 2), in which case $\mathcal{E}^{\text{gwc}} = \mathbb{R}^d_+$. Nonetheless, our construction remains well-defined even in that special case.

Using Lemma 7, we compute $S^i_{\text{lwc}}(\mathcal{Y}^h) = \hat{\Phi}^{\text{lwc}}_h(\mathbf{E}^i)$ for all $i \in [n]$ and define $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h) := \hat{Q}_{1-\alpha}(S^1_{\text{lwc}}(\mathcal{Y}^h), \ldots, S^n_{\text{lwc}}(\mathcal{Y}^h))$. Under the working hypothesis $\mathbf{Y}^{n+1} \in \mathcal{Y}^h$, it follows immediately from (23) that $\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h) \geq \hat{Q}^{\text{oracle}}_{1-\alpha}$ almost surely. By substituting $c = \hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h)$ in (11), we obtain the local prediction set $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)$ in (18) with $\mathcal{R} = \mathcal{Y}^h$ and it satisfies (19).

The procedure above summarizes the construction when $\mathcal{Y}^h \neq \emptyset$. If $\mathcal{Y}^h = \emptyset$, we can simply set $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h) = \emptyset$ and the containment (19) holds trivially. Algorithm 1 summarizes this construction.

Algorithm 1 Construction of local prediction sets under working hypothesis $\mathbf{Y}^{n+1} \in \mathcal{Y}^h$

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; global bounds $\omega_1(\hat{Q}^{\text{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\text{gwc}}_{1-\alpha})$; calibration mean $\hat{\mu}^{\text{obs}}$ and standard deviation $\hat{\sigma}^{\text{obs}}$; index $h \in [n+1]^d$.

```
1: if Y^h ≠ ∅ then
2:     Compute S^i_lwc(Y^h) ← Φ̂^lwc_h(E^i) for i ∈ [n] using Φ̂^lwc_h defined in Lemma 7.
3:     Compute Q̂^lwc_{1−α}(Y^h) ← Q̂_{1−α}(S^1_lwc(Y^h), ..., S^n_lwc(Y^h)).
4:     return ω_1(Q̂^lwc_{1−α}(Y^h)), ..., ω_d(Q̂^lwc_{1−α}(Y^h)) using ω defined in Lemma 2.
5: else
6:     return ∅.
7: end if
```

3.4.4 Computational Shortcut for Rectangular Enclosure

While the characterization provided above makes it easy to compute $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)$ for any specific index $h \in [n+1]^d$, the problem remains that the intended construction of $\hat{C}^{\text{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})$ in (20) involves an *exponential* number of such local prediction sets.

Therefore, to obtain a practical method, we must now develop a shortcut that avoids full enumeration of all $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)$. Our solution begins by noting that $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)$ from (18) can be re-written as:

$$\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h) = \bigcap_{j=1}^{d} \left\{ \mathbf{y} \in \mathbb{R}^d : E_j^{(h_j - 1)} \leq E_j(\mathbf{X}^{n+1}, y_j) \leq B_j^h \right\} \subseteq \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}),$$

where the coordinate-wise upper bounds B_j^h are given by

$$B_j^h := \begin{cases} \min\left\{ U_j^{h_j},\ \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h)) \right\}, & \text{if } U_j^{h_j} > E_j^{(h_j-1)} \text{ and } \omega_j(\hat{Q}^{\text{lwc}}_{1-\alpha}(\mathcal{Y}^h)) > E_j^{(h_j-1)}, \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

For every coordinate $j \in [d]$, define the j-th residual boundary $\mathcal{L}_j := \max_{h \in [n+1]^d} B_j^h$. Then the rectangular enclosure of all local prediction sets in (20) can be expressed as

$$\begin{aligned}\mathrm{Rect}_{\mathcal{P}_{n,d}}\left(\bigsqcup_{h\in[n+1]^d} \hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{Y}^h)\right) &= \bigcap_{j=1}^{d} \left\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \mathcal{L}_j\right\} \\ &=: \hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1}).\end{aligned} \tag{25}$$

Notably, this rectangular enclosure is typically very tight in practice, because the union of the local prediction sets constructed using our designated partition $\mathcal{P}_{n,d}$ is itself very close to being a rectangle when the sample size is large enough.

The last remaining step to make our method practical is finding an efficient algorithm for computing the boundaries $\mathcal{L}_j$ for all $j \in [d]$, without requiring enumeration of all local prediction sets. Such a shortcut is made possible by the following result, which simplifies the definition of $\mathcal{L}_j$ from a maximum over $(n+1)^d$ distinct values to a maximum over an explicit, narrowed-down list of only $n+1$ relevant candidates.

Before stating this result, it is helpful to introduce some notation. Define the (random) *mean index* vector $h^* \in [n+1]^d$ as the unique index such that $E_j^{(h_j^*-1)} \le \hat{\mu}_j^{\mathrm{obs}} \le E_j^{(h_j^*)}$ for all $j \in [d]$. Let $\mathrm{Row}_j(h^*)$ be a subset of $[n+1]^d$ containing exactly $n+1$ indices, defined as those coinciding with the mean index h^* on all coordinates other than the j-th one:

$$\mathrm{Row}_j(h^*) := \left\{h \in [n{+}1]^d : h_k = h_k^* \text{ for all } k \neq j\right\}, \quad \forall j \in [d].$$

Lemma 8 *If $\mathcal{Y}^{h^*} \neq \emptyset$, then for any $j \in [d]$, then $\mathcal{L}_j := \max_{h \in [n+1]^d} B_j^h$ can be equivalently written as:*

$$\mathcal{L}_j = \max_{h \in \mathrm{Row}_j(h^*)} B_j^h = B_j^{h^{[j]}} > 0, \quad \textit{where } h^{[j]} := \operatorname*{argmax}_{h \in \mathrm{Row}_j(h^*)} B_j^h.$$

This maximum index $h^{[j]}$ satisfies $B_j^h = 0$ for any $h \in \mathrm{Row}_j(h^*)$ with $h_j \ge h_j^{[j]}$.

Moreover, for every $j \in [d]$, if there exists $m \in \mathrm{Row}_j(h^*)$ with $m_j \ge h_j^*$ and $B_j^m = 0$, then $B_j^h = 0$ for all $h \in \mathrm{Row}_j(h^*)$ with $h_j \ge m_j$.

Under the regularity condition $\mathcal{Y}^{h^*} \neq \emptyset$, which can be immediately verified and typically holds in practice under all but the most extreme scenarios, Lemma 8 gives a very fast algorithm for computing $\mathcal{L}_j$ separately for each coordinate $j \in [d]$. In particular, one can find $h^{[j]}$, where only $h_j^{[j]}$ needs to be determined, by performing a simple backward search over $h \in \mathrm{Row}_j(h^*)$ with $h_j = n+1, \ldots, 1$ and stopping for the first h_j with $B_j^h > 0$. Moreover, if $h_j^{[j]} \ge h_j^*$, which can be determined by simply checking whether $B_j^{h^*} > 0$, we can find $h^{[j]}$ even more effectively through a binary search that checks whether $B_j^m = 0$ at each intermediate split step $m \in \mathrm{Row}_j(h^*)$ with $m_j \in [h_j^*, ..., n+1]$ and assigns $h_j^{[j]}$ to be the largest m_j such that $B_j^m > 0$.

The special case $\mathcal{Y}^{h^*} = \emptyset$ corresponds to an extreme situation where even the conservative GWC prediction set $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ does not contain the empirical mean $\hat{\boldsymbol{\mu}}^{\mathrm{obs}}$ of the calibration residuals. Typically, this should not occur at moderate miscoverage levels α, unless the data are extremely heavy-tailed. Such cases can be handled by simply falling back to the conservative GWC prediction set $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ itself, which we know to have valid marginal coverage. In conclusion, the procedure described above, summarized in Algorithm 2, produces:

$$\hat{C}(\mathbf{X}^{n+1}) = \begin{cases} \hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1}), & \text{if } \mathcal{Y}^{h^*} \neq \emptyset, \\ \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1}), & \text{otherwise.} \end{cases} \tag{26}$$

Theorem 9 *Assume* $(\mathbf{E}^1, \ldots, \mathbf{E}^{n+1})$ *are exchangeable and Assumption 1 holds. Then,* $\hat{C}(\mathbf{X}^{n+1})$ *constructed by Algorithm 2 satisfies:* $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}(\mathbf{X}^{n+1})] \geq 1 - \alpha$.

Proof Using the characterization of $\hat{C}(\mathbf{X}^{n+1})$ in (26), the proof follows immediately by combining Proposition 6, which tells us that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{lwc}}_{\mathcal{P}_{n,d}}(\mathbf{X}^{n+1})] \geq 1-\alpha$, and Theorem 5, which tells us that $\mathbb{P}[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})] \geq 1 - \alpha$. ■

Theorem 9 guarantees that our prediction sets $\hat{C}(\mathbf{X}^{n+1})$ defined in (26) have valid finite-sample marginal coverage above the target level $1 - \alpha$. Moreover, these prediction sets have an interpretable rectangular shape and are typically much tighter than $\hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$.

Algorithm 2 Transductively Standardized Conformal Prediction (TSCP)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

Compute $\hat{\mu}, \hat{\sigma}$ using (14).
Compute global bounds $\omega_1(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \ldots, \omega_d(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$ using TSCP-GWC. ▷ Algorithm S5.
Sort $\{E^i_j\}_{i=1}^n$ for each $j \in [d]$ to obtain order statistics $E_j^{(k)}$.
Compute $\mathcal{Y}^{h^*}$ using (21).
if $\mathcal{Y}^{h^*} = \emptyset$ **then**
  return $\hat{C}(\mathbf{X}^{n+1}) \leftarrow \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \quad \forall j \in [d] \right\}$.
else
  for $j = 1$ to d **do**
    Compute $\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^{h^*}))$ using Algorithm 1 and $B_j^{h^*}$ using (24).
    if $B_j^{h^*} = 0$ **then**
      Backward search over $h_j = h_j^* - 1, \ldots, 1$ to find $h^{[j]}$. ▷ Algorithm S6
    else
      Binary search over $h_j = h_j^*, \ldots, n{+}1$ to find $h^{[j]}$. ▷ Algorithm S7
    end if
    Update $\mathcal{L}_j \leftarrow B_j^{h^{[j]}}$.
  end for
  return $\hat{C}(\mathbf{X}^{n+1}) \leftarrow \left\{ \mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \mathcal{L}_j, \quad \forall j \in [d] \right\}$.
end if

The computational complexity of Algorithm 2 ranges from $\mathcal{O}(d^2 n \log n)$ in the best case, when binary searches can be used for all coordinates, to $\mathcal{O}(d^2n^2)$ in the worst case, when all coordinates require backward searches; see Appendix A.3 for details. Even in the worst case, a cost of $\mathcal{O}(d^2n^2)$ is typically not prohibitive in practice: calibration sample sizes n

for conformal prediction are usually in the hundreds to low thousands, and beyond that additional samples are often better spent improving model training.

In addition, the computation of the distinct boundaries $\mathcal{L}_1, \ldots, \mathcal{L}_d$ is trivially parallelizable, and the entire procedure need not be repeated across different test inputs, since $\mathcal{L}_1, \ldots, \mathcal{L}_d$ are independent of $\mathbf{X}^{n+1}$—remarkably, we have arrived through *transductive* logic at a method that is effectively *inductive*. Therefore, applying Algorithm 2 to m different test points using the same calibration data set has total cost ranging from $\mathcal{O}(d^2 n \log n + dm)$ to $\mathcal{O}(d^2 n^2 + dm)$ for typical choices of generalized residuals (Section 2.3).

4 Numerical Experiments

4.1 Methods and Metrics

We provide a comprehensive empirical validation of the proposed *TSCP* method (Algorithm 2) using simulated and real data. The performance of TSCP is compared to that of three representative benchmark approaches: (i) *Unscaled Max*, a direct extension of split conformal inference that applies the ℓ_∞ norm to the residuals, summarized in Algorithm S2; (ii) *Point CHR*, a data-splitting variant recently proposed by Sampson and Chan (2024), which estimates per-target out-of-sample points using a held-out calibration set and adjusts them to form axis-aligned rectangular sets; (iii) *Emp. Copula*, the first distribution-free copula-based method introduced in Messoudi et al. (2021).

The results of additional experiments comparing TSCP to different approaches are in Appendix C. We do not include in this empirical study alternative copula-based methods—e.g., vine copulas (Park and Cho, 2025)—because their main limitations relative to our approach are the same as those of the *Emp. Copula* benchmark. To enable meaningful direct comparisons, we also omit approaches that do not produce rectangular prediction sets.

For simplicity, we apply all methods using the same conditional mean-regression model $\hat{f}$ and standard absolute residuals; i.e., $E_j(\mathbf{X}, Y_j) = |Y_j - \hat{f}_j(\mathbf{X})|$ for all $j \in [d]$. Although our method can also be seamlessly applied using different choices of models and residuals, this is a relatively common choice previously employed in several related works (Sampson and Chan, 2024; Park and Cho, 2025; Dheur et al., 2025).

The conformal prediction sets produced by the different methods are compared based on two key performance metrics: average coverage and size (volume). Since all methods rely on the same underlying model, it suffices to measure the size of the prediction sets in residual space. Specifically, we report

$$\text{(Test) Coverage} := \frac{1}{N} \sum_{l=1}^{N} \frac{\#\text{ of included test points}}{\#\text{ of total test points}},$$

$$\text{(Residual space) Volume} := \frac{1}{N} \sum_{l=1}^{N} \frac{1}{2^d} \text{Vol(prediction set)},$$

where both metrics are averaged over N repetitions of each experiment, corresponding to independent data sets. The constant factor 2^{-d} in the volume definition facilitates comparison across experiments with different output dimensions.

4.2 Experiments with Simulated Data

We simulate data with features from a 10-dimensional standard normal distribution with independent components, and 10-dimensional outcomes

$$Y_j \mid X \sim f_j(X) + \epsilon_j, \quad \forall j \in [10], \tag{27}$$

where $f_j(X) = \sum_{i=1}^{10} \xi_i X_i$ such that $\xi_1, \ldots, \xi_{10} \sim \mathrm{Unif}(-10, 10)$, and ϵ_j is independent random noise for each $j \in [d]$. We consider six settings corresponding to different noise distributions; the results for two are presented here while those corresponding to other settings, which lead to qualitatively similar conclusions, are summarized in Appendix C.1.

In each setting, we simulate a training data set containing 7200 observations to fit a linear regression model $\hat{f}$ using ordinary least squares, a calibration set $\mathcal{D}_{\text{cal}}$ of size $|\mathcal{D}_{\text{cal}}| \in \{30, 50, 100, 300, 500\}$, and a test set $\mathcal{D}_{\text{test}}$ of size 800, to construct prediction sets at level $\alpha = 0.1$. All results are averaged over $N = 200$ independent repetitions.

Figure 3 summarizes the results obtained in the first setting, where the noise distribution is Gaussian with heterogeneous variance across dimensions:

$$\epsilon_j^{\text{heter}} \sim \mathcal{N}\left(0,\ (10 - j + 1)^2\right), \quad \forall j \in [10].$$

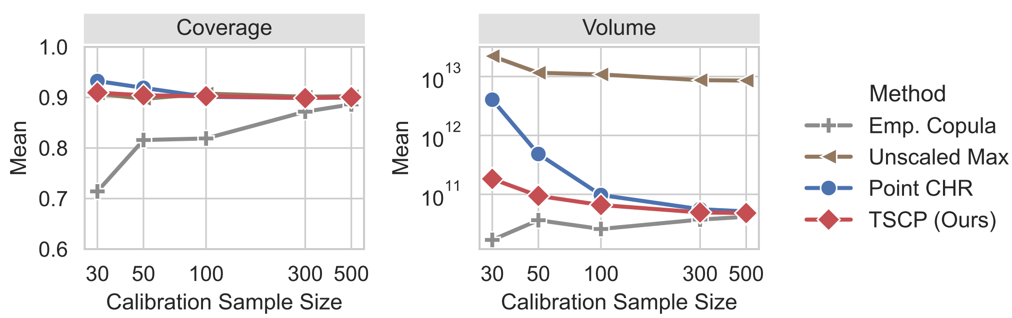

Figure 3: Average performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables and heterogeneous noise variances. The target coverage level is 90%. TSCP consistently attains near-nominal coverage while producing the smallest prediction sets on average. Numerical results and corresponding standard deviations are reported in Table S3.

These results show that our method performs well across both small and large sample sizes, always achieving valid coverage at level $1 - \alpha$ (as predicted by the theory) but not excessively above it, using relatively tight prediction sets. The *Point CHR* approach yields substantially larger, and therefore less informative, prediction sets—especially in small samples (e.g., $n < 100$)—due to the inefficiency introduced by its additional sample split. Even for moderately large samples, our method continues to perform noticeably better, although this difference is difficult to discern in Figure 3 because of the log–log scale; see Table S3 for a more detailed numerical summary. The *Emp. Copula* approach clearly fails to achieve finite-sample coverage, since it uses the same calibration set both to estimate the copula model and to calibrate the prediction sets based on the resulting nonconformity scores. While this issue could be addressed through additional data splitting, doing so would come with the same efficiency disadvantages of *Point CHR*. This limitation is shared by related copula-based approaches that we do not explicitly include in our comparisons. Finally, the *Unscaled Max* approach produces much larger (and effectively uninformative) prediction sets in both small and large samples, as it is negatively affected by the unequal variances of the noise distribution across different dimensions.

Figure 4 reports results from analogous experiments in which the noise distribution has equal variance across dimensions, with $\epsilon_j^{\text{homo}} \sim \mathcal{N}(0,1)$ for all $j \in [10]$. This setting strongly favors the *Unscaled Max* approach; nevertheless, our TSCP still performs as a close second. In practice, however, noise levels are rarely known *a priori*, making the adaptivity and robustness of TSCP generally desirable.

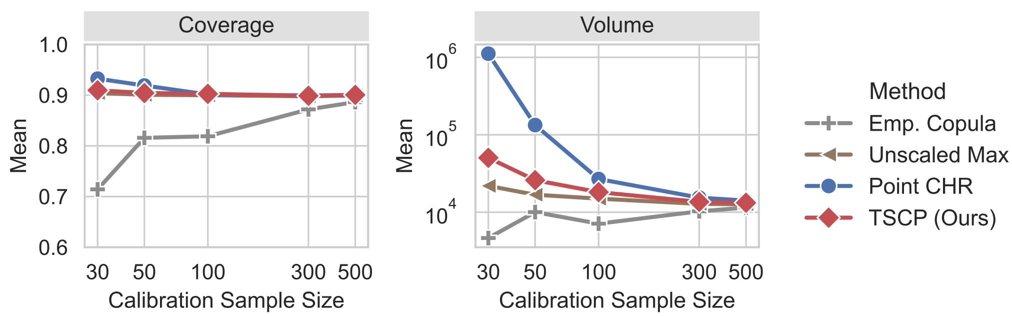

Figure 4: Average performance of the proposed TSCP method and alternative conformal prediction approaches in numerical experiments similar to those of Figure 3. Here, the outcome variables have homogeneous variance across their $d = 10$ dimensions, which gives an advantage to the *Unscaled Max* approach. The target coverage level is 0.9. TSCP consistently attains near-nominal coverage while producing the smallest prediction sets on average. Numerical results and corresponding standard deviations are reported in Table S4.

The results of additional numerical experiments are reported in Appendices C.2–C.3. In Appendix C.2, we compare TSCP with several alternative approaches described in Sections 2 and 3, including the naive plug-in rescaling procedure (Algorithm S3), the data-splitting variant (Algorithm S4), the conservative global worst-case approach (Algorithm S5), and the union-based construction without rectangular enclosure (Section 3.4). These experiments highlight the favorable balance between statistical–computational trade-offs achieved by TSCP. We further examine scalability with respect to the outcome dimension and show that Algorithm 2 remains computationally efficient in regimes where union-based constructions become prohibitive. In Appendix C.3, we investigate the robustness of TSCP under heavy-tailed noise and demonstrate that our method continues to achieve near-nominal coverage even in extreme settings where the population variance of the residuals is infinite.

4.3 Experiments with Real Data

We now evaluate all methods on six real datasets. The first four, *rf2* (Tsoumakas et al., 2022), *scm1d* (Contributors, 2019a), *scm20d* (Contributors, 2019b), and *stock* (Yeh, 2015), are borrowed from related prior works (Park and Cho, 2025; Dheur et al., 2025). The other two datasets, *student* (Cortez, 2008) and *energy* (Tsanas and Xifara, 2012), are obtained from the UCI repository and serve to illustrate the small-sample performance of TSCP.

For each dataset, we perform 200 random train/calibration/test splits using a 75%/5%/20% ratio. We fit a multivariate sparse (lasso) regression model with regularization parameter $\lambda = 0.0001$ on *stock*, and random forest regression models on all other datasets; see the GitHub software repository linked in the Software Availability section for additional reproducibility details. Table 1 reports the average coverage and volume, along with one standard deviation. As shown in Table 1, the empirical behavior of all methods on real data largely mirrors the trends observed in the simulation studies. Among methods with theoretical coverage guarantees, TSCP achieves the smallest prediction-set volume on the *stock*, *scm1d*, *scm20d*, and *energy* datasets. On *rf2*, TSCP is outperformed by *Point CHR*, likely due to the presence of a small number of influential outliers. On *student*, *Unscaled Max* yields smaller volumes, consistent with the approximately homogeneous noise across targets (student GPAs at different grade levels). The *Point CHR* approach produces an uninformative, *infinite* prediction set on the *stock* dataset, because its additional data split makes the effective calibration set extremely small. Finally, *Emp. Copula* generally fails to achieve the desired coverage level, consistent with its lack of finite-sample guarantees. Overall, the results show that TSCP provides a reasonable trade-off between coverage and volume, excelling especially when calibration data are limited or when noise scales differ significantly across targets.

To emphasize their interpretability, Table 2 displays the rectangular prediction sets produced by different methods for two representative test points from the *energy* dataset, where $d = 2$. In this setting, rectangular prediction sets consist of pairs of simultaneous prediction intervals that aim to jointly contain the true outcomes with probability 90%.

5 Discussion and Future Work

The core idea underlying the method presented in this paper is the explicit standardization of residuals, which contributes both to the transparency and to the computational efficiency of the resulting prediction sets. As with other standardization-based methods, however, this strategy introduces some sensitivity to outliers. This behavior is visible in the numerical experiments based on the *rf2* dataset, where a small number of influential observations can substantially expand the resulting prediction sets. An important direction for future work is therefore the development of robust extensions of our method, for example through rescaling based on median absolute deviation or quantile-based normalizations, while preserving the exchangeability required for conformal validity.

Additional directions for future work include allowing for negative residuals. Such an extension could, for example, enable shrinkage of baseline prediction sets when starting from quantile regression models that already exhibit conservative coverage. It would also be of interest to adapt the proposed approach to settings with non-numerical outcomes, such as categorical or ordinal responses. Finally, our method could be extended to handle partly observed outcomes (Braun et al., 2025b).

Method	Stock ($d = 6$, $n = 15$)		rf2 ($d = 8$, $n = 384$)	
	Coverage	Volume	Coverage	Volume
Unscaled Max	0.940(0.057)	6.03e-02(1.57e-01)	0.900(0.016)	4.14e+03(2.90e+03)
Emp. copula	0.727(0.113)	1.01e-05(9.80e-06)	0.893(0.016)	5.35e+01(4.69e+01)
Point CHR	1.000(0.000)	inf	0.901(0.022)	**6.85e+01(7.55e+01)**
TSCP	0.956(0.059)	**1.86e-03(8.55e-03)**	0.901(0.017)	5.42e+02(1.50e+03)

Method	scm1d ($d = 16$, $n = 490$)		scm20d ($d = 16$, $n = 448$)	
	Coverage	Volume	Coverage	Volume
Unscaled Max	0.900(0.016)	2.74e+39(4.02e+39)	0.901(0.016)	2.70e+40(2.92e+40)
Emp. copula	0.893(0.016)	1.09e+39(1.34e+39)	0.892(0.017)	1.178e+40(1.190e+40)
Point CHR	0.905(0.020)	2.84e+39(5.43e+39)	0.902(0.023)	3.00e+40(8.00e+40)
TSCP	0.901(0.015)	**1.52e+39(2.61e+39)**	0.901(0.016)	**1.53e+40(1.50e+40)**

Method	Energy ($d = 2$, $n = 38$)		Student ($d = 3$, $n = 32$)	
	Coverage	Volume	Coverage	Volume
Unscaled Max	0.917(0.049)	1.58e+01(6.80e+00)	0.904(0.055)	**1.51e+02(1.27e+02)**
Emp. copula	0.886(0.061)	5.41e+00(4.49e+00)	0.861(0.073)	1.08e+02(7.19e+01)
Point CHR	0.899(0.069)	8.81e+00(2.42e+01)	0.939(0.057)	6.89e+02(1.07e+03)
TSCP	0.922(0.048)	**6.95e+00(4.43e+00)**	0.912(0.056)	1.98e+02(1.77e+02)

Table 1: Performance of the proposed TSCP method and alternative conformal prediction approaches on real datasets. Results are averaged over 200 random splits of the data into training/calibration/and test subsets, with one standard deviation reported in parentheses. The target coverage level is 0.9. Coverage values below the target are highlighted in red, and the smallest volume among methods achieving valid coverage is highlighted in **bold**.

Test 1/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[31.70, 38.94]	**[34.74, 41.98]**	13.09	(35.40, 39.22)
Emp. copula	**[34.36, 36.28]**	**[34.74, 41.98]**	3.47	
Point CHR	[34.29, 36.36]	[31.45, 45.27]	7.14	
TSCP	[34.28, 36.36]	[33.63, 43.08]	4.93	

Test 2/Methods	Outcome 1	Outcome 2	Volume	Ground Truth
Unscaled Max	[25.44, 32.67]	[29.34, 36.58]	13.09	(29.88, 28.31)
Emp. copula	**[28.10, 30.02]**	[29.34, 36.58]	3.47	
Point CHR	[28.02, 30.09]	[26.05, 39.87]	7.14	
TSCP	[28.01, 30.10]	**[28.24, 37.69]**	4.93	

Table 2: Anecdotal illustration of conformal prediction sets of rectangular shape for two test points from the *energy* dataset. In this case, the prediction sets can be reported as pairs of joint prediction intervals for the two outcome dimensions. Volumes are computed in residual space. Intervals that fail to cover the corresponding true outcomes are highlighted in red; smallest interval that covers the ground truth is highlighted in **bold**. This example is included for illustration only: the coverage guarantees of conformal prediction hold on average, not for individual test points.

Software Availability

A software implementation of the methods described in this paper, along with code necessary to reproduce the numerical experiments, is available online at `https://github.com/OTheMagic/multi-target-scaling.git`.

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Appendix: Algorithms, Proofs, and Numerical Results

Appendix A. Additional Algorithmic Details

A.1 Implementation of Alternative Methods

Algorithm S1 Separate Prediction Intervals with Bonferroni Correction

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

Compute $\alpha^* \leftarrow \alpha/d$.
for $j = 1$ to d **do**
  Compute $\hat{Q}_{1-\alpha^*,j} \leftarrow \hat{Q}_{1-\alpha^*}(E_j^1, \ldots, E_j^n)$.
end for
return $\hat{C}^{\text{Bonf.}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}_{1-\alpha^*,j}, \forall j \in [d]\right\}$.

Algorithm S2 Prediction Intervals via Dimension Reduction with ℓ_∞-Norm (Unscaled)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

Compute $S^i_{\text{unscaled}} \leftarrow \|\mathbf{E}^i\|_\infty$ for $i = 1, \ldots, n$.
Compute $\hat{Q}^{\text{unscaled}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{unscaled}}, \ldots, S^n_{\text{unscaled}})$
return $\hat{C}^{\text{unscaled}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{unscaled}}_{1-\alpha}, \forall j \in [d]\right\}$.

Algorithm S3 Heuristic Plug-In Standardization

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

for $j = 1$ to d **do**
  Compute the location and scale parameters

$$\hat{\mu}_j^{\text{pi}} \leftarrow \frac{1}{n}\sum_{i=1}^n E_j^i, \qquad \hat{\sigma}_j^{\text{pi}} \leftarrow \sqrt{\frac{1}{n-1}\sum_{i=1}^n (E_j^i - \hat{\mu}_j^{\text{pi}})^2}.$$

end for
Compute $S^i_{\text{pi}} \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{pi}}, \hat{\boldsymbol{\sigma}}^{\text{pi}})$ for $i \in [n]$, using Φ defined in (6).
Compute $\hat{Q}^{\text{pi}}_{1-\alpha} \leftarrow \hat{Q}_{1-\alpha}(S^1_{\text{pi}}, \ldots, S^n_{\text{pi}})$.
Compute

$$\hat{C}^{\text{pi}}(\mathbf{X}^{n+1}) \leftarrow \left\{\mathbf{y} \in \mathbb{R}^d : 0 \leq E_j(\mathbf{X}^{n+1}, y_j) \leq \hat{Q}^{\text{pi}}_{1-\alpha} \cdot \hat{\sigma}_j^{\text{pi}} + \hat{\mu}_j^{\text{pi}}, \forall j \in [d]\right\},$$

which is hyper-rectangular for the standard choices of $\mathbf{E}$ discussed in Section 2.4.
return Prediction set $\hat{C}^{\text{pi}}(\mathbf{X}^{n+1})$.

Algorithm S4 Standardization with Data Splitting

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.
Output: $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1})$

Randomly split $\{1,\dots,n\}$ into two disjoint subsets, $\mathcal{I}_1$ and $\mathcal{I}_2$.
for $j = 1$ to d **do**
  Compute the location and scale parameters

$$\hat{\mu}_j^{\text{ds}} \leftarrow \frac{1}{|\mathcal{I}_1|}\sum_{i\in\mathcal{I}_1} E_j^i, \qquad \hat{\sigma}_j^{\text{ds}} \leftarrow \sqrt{\frac{1}{|\mathcal{I}_1|-1}\sum_{i\in\mathcal{I}_1}(E_j^i - \hat{\mu}_j^{\text{ds}})^2}.$$

end for
Compute $S_{\text{ds}}^i \leftarrow \Phi(\mathbf{E}^i; \hat{\boldsymbol{\mu}}^{\text{ds}}, \hat{\boldsymbol{\sigma}}^{\text{ds}})$ for $i \in \mathcal{I}_2$, using Φ defined in (6).
Compute $\hat{Q}_{1-\alpha}^{\text{ds}} \leftarrow \hat{Q}_{1-\alpha}(S_{\text{ds}}^1, \dots, S_{\text{ds}}^n)$.
return $\hat{C}^{\text{ds}}(\mathbf{X}^{n+1}) \leftarrow \Big\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \hat{Q}_{1-\alpha}^{\text{ds}} \cdot \hat{\sigma}_j^{\text{ds}} + \hat{\mu}_j^{\text{ds}}, \forall j \in [d]\Big\}$.

A.2 Auxiliary Algorithms

Algorithm S5 Global-Worst-Case TSCP (TSCP-GWC)

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; target miscoverage level $\alpha \in (0,1)$; test input $\mathbf{X}^{n+1}$.

Compute $\hat{\boldsymbol{\mu}}^{\text{obs}}, \hat{\boldsymbol{\sigma}}^{\text{obs}}$ using (14).
Compute $S_{\text{gwc}}^i \leftarrow \hat{\Phi}^{\text{gwc}}(\mathbf{E}^i)$ using $\hat{\Phi}^{\text{gwc}}$ defined in Lemma 4.
Compute $\hat{Q}_{1-\alpha}^{\text{gwc}} \leftarrow \hat{Q}_{1-\alpha}(S_{\text{gwc}}^1, \dots, S_{\text{gwc}}^n)$.
Compute $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \dots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$ using ω defined in Lemma 2.
return $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \leftarrow \Big\{\mathbf{y} \in \mathbb{R}^d : 0 \le E_j(\mathbf{X}^{n+1}, y_j) \le \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}}), \forall j \in [d]\Big\}$.

Algorithm S6 Backward Search in Algorithm 2

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}_{1-\alpha}^{\text{gwc}}), \dots, \omega_d(\hat{Q}_{1-\alpha}^{\text{gwc}})$; target miscoverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\text{obs}}$; mean index h^*; current coordinate j.

for $h \in \text{Row}_j(h^*)$ with $h_j = h_j^* - 1$ to 1 **do**
  Compute $\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h))$ using Algorithm 1 and B_j^h using (24).
  if $B_j^h > 0$ **then**
    return $h^{[j]} = h$.
  end if
end for

Algorithm S7 Binary Search in Algorithm 2

Input: Calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$; global bounds $\omega_1(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$; target mis-coverage level $\alpha \in (0,1)$; calibration mean $\hat{\boldsymbol{\mu}}^{\mathrm{obs}}$ and standard deviation $\hat{\boldsymbol{\sigma}}^{\mathrm{obs}}$; mean index h^*; current coordinate j.

Let $I_{\min} \leftarrow h^*_j$ and $I_{\max} \leftarrow n+1$.
while $I_{\min} < I_{\max}$ **do**
  Compute $m_j \leftarrow \lceil (I_{\min} + I_{\max})/2 \rceil$ and $m_k = h^*_k$ for all $k \neq j$.
  Compute $\omega_j(\hat{Q}^{\mathrm{lwc}}_{1-\alpha}(\mathcal{Y}^m))$ and B^m_j using (24). ▷ Algorithm 1
  if $B^m_j > 0$ **then**
    $I_{\min} \leftarrow m_j$.
  else
    $I_{\max} \leftarrow m_j - 1$.
  end if
  if $I_{\max} = I_{\min}$ **then**
    return $h^{[j]}_j = I_{\min}$ and $h^{[j]}_k = h^*_k$ for all $k \neq j$.
  end if
end while

A.3 Computational Cost Analysis

A.3.1 Algorithm S5

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n,d)-array, we know

1. Computing $(\hat{\mu}^{\mathrm{obs}}_j, \hat{\sigma}^{\mathrm{obs}}_j)_{j=1}^d$ using (14) requires $\mathcal{O}(nd)$.
2. Evaluating $S^i_{\mathrm{gwc}} \leftarrow \hat{\Phi}^{\mathrm{gwc}}(\mathbf{E}^i)$ for all $i \in [n]$ is also $\mathcal{O}(nd)$.
3. Computing the quantile $\hat{Q}^{\mathrm{gwc}}_{1-\alpha}$ costs $\mathcal{O}(n)$ using a selection algorithm such as quickselect.
4. Computing $\omega_1(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$ costs $\mathcal{O}(d)$ once $\hat{Q}^{\mathrm{gwc}}_{1-\alpha}$ and $(\hat{\mu}^{\mathrm{obs}}_j, \hat{\sigma}^{\mathrm{obs}}_j)_{j=1}^d$ are known.

Hence, the overall computational complexity of Algorithm S5 is $\mathcal{O}(nd)$.

A.3.2 Algorithm 1

Algorithm 1 shares the same structure and computational cost as Algorithm S5: $\mathcal{O}(nd)$.

A.3.3 Algorithm 2

Since the calibration residuals $\{\mathbf{E}^i\}_{i=1}^n$ are stored as an (n,d)-array, we know

1. Computing $(\hat{\mu}^{\mathrm{obs}}_j, \hat{\sigma}^{\mathrm{obs}}_j)_{j=1}^d$ using (14) and $\omega_1(\hat{Q}^{\mathrm{gwc}}_{1-\alpha}), \dots, \omega_d(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})$ using Algorithm S5 requires $\mathcal{O}(nd)$.
2. Sorting $\{E^i_j\}_{i=1}^n$ for all $j \in [d]$ costs $\mathcal{O}(dn \log n)$;

3. Computing $\mathcal{Y}^{h^*}$ relies on computing h^*, which requires masking all values below $\hat{\boldsymbol{\mu}}^{\text{obs}}$ and reporting the index of the lowest value from the sorted residuals list $\{E_j^i\}_{i=1}^n$, each of which costs $\mathcal{O}(n)$, so together this operation contributes $\mathcal{O}(dn)$.
4. If $\mathcal{Y}^{h^*} = \emptyset$, then the cost is only $\mathcal{O}(d)$ since all global bounds have been computed.
5. If $\mathcal{Y}^{h^*} \neq \emptyset$, then for each coordinate, we compute the local bounds using Algorithm 1 and $B_j^{h^*}$ using (24), which costs $\mathcal{O}(dn)$. If $B_j^{h^*} = 0$, then we use the backward search detailed in Algorithm S6, which costs $\mathcal{O}(dn^2)$; see Appendix A.3.4. If $B_j^{h^*} > 0$, then we use the binary search detailed in Algorithm S7, which costs $\mathcal{O}(dn \log n)$; see Appendix A.3.5.

Let $T \leq d$ be the number of coordinates where binary searches are enabled, then the total cost of Algorithm 2 is $\mathcal{O}(Tdn \log n + (d-T)dn^2)$.

A.3.4 Algorithm S6

Running Algorithm 1 and forming B_j^h costs $\mathcal{O}(dn)$ for each h; there are $n+1$ indices in $\text{Row}_j(h^*)$, so the computational complexity of Algorithm S6 never exceeds $\mathcal{O}(dn^2)$.

A.3.5 Algorithm S7

Running Algorithm 1 and forming B_j^h costs $\mathcal{O}(dn)$ for each h. The number of indices h encountered in this algorithm is $\mathcal{O}(\log_2 n)$ because we split and continue in only half of the current indices sequence at each step. This is a standard binary search. Therefore, the total computational complexity of Algorithm S7 never exceeds $\mathcal{O}(dn \log n)$.

Appendix B. Mathematical Proofs

B.1 Auxiliary Lemmas

Lemma S1 *Let $x^1, \dots, x^n \in \mathbb{R}$ and $\bar{x}^n = \frac{1}{n}\sum_{i=1}^n x^i$. For any $x^{n+1} \in \mathbb{R}$, let $\hat{\sigma}(x^{n+1})$ be the sample standard deviation of $\{x^1, \dots, x^n, x^{n+1}\}$, then*

$$\hat{\sigma}^2(x^{n+1}) = \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.$$

Proof Denote $\bar{x}^{n+1} = \frac{1}{n+1}\sum_{i=1}^{n+1} x^i$. We notice that

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\sum_{i=1}^{n+1}(x^i - \bar{x}^{n+1})^2 \\
=& \frac{1}{n}\left[(x^{n+1} - \bar{x}^{n+1})^2 + \sum_{i=1}^n (x^i - \bar{x}^n + \bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + \frac{1}{n}\sum_{i=1}^n \left[(x^i - \bar{x}^n)^2 + 2(x^i - \bar{x}^n)(\bar{x}^n - \bar{x}^{n+1}) + (\bar{x}^n - \bar{x}^{n+1})^2\right] \\
=& \frac{(x^{n+1} - \bar{x}^{n+1})^2}{n} + (\bar{x}^n - \bar{x}^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

Substituting $\bar{x}^{n+1} = \frac{x^{n+1} + n\bar{x}^n}{n+1}$, we get

$$\begin{aligned}
\hat{\sigma}^2(x^{n+1}) =& \frac{1}{n}\left(x^{n+1} - \frac{x^{n+1} + n\bar{x}^n}{n+1}\right)^2 + \left(\bar{x}^n - \frac{n\bar{x}^n + x^{n+1}}{n+1}\right)^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{n}{(n+1)^2}(x^{n+1} - \bar{x}^n)^2 + \frac{1}{(n+1)^2}(\bar{x}^n - x^{n+1})^2 + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2 \\
=& \frac{(x^{n+1} - \bar{x}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (x^i - \bar{x}^n)^2.
\end{aligned}$$

■

Lemma S2 *Let $X^1, \dots, X^n \in \mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^n$. Let X^{n+1} and $\tilde{X}^{n+1}$ be another two random variables such that*

$$|X^{n+1} - \bar{X}^n| \leq |\tilde{X}^{n+1} - \bar{X}^n| \quad a.s.$$

Then,

$$\hat{\sigma}(X^{n+1}) \leq \hat{\sigma}(\tilde{X}^{n+1}) \quad a.s.,$$

where $\hat{\sigma}(X^{n+1})$ is the sample standard deviation of $\{X^1, \dots, X^n, X^{n+1}\}$, as in Lemma S1.

Proof This is a direct application of Lemma S1:

$$\begin{aligned}\hat{\sigma}^2(X^{n+1}) &= \frac{(X^{n+1}-\bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (X^i - \bar{X}^n)^2 \\ &\le \frac{(\tilde{X}^{n+1}-\bar{X}^n)^2}{n+1} + \frac{1}{n}\sum_{i=1}^n (X^i - \bar{X}^n)^2 \quad \text{[Assumption]} \\ &= \hat{\sigma}^2(\tilde{X}^{n+1}),\end{aligned}$$

and the claim follows by taking square roots. ■

Lemma S3 *Let $X^1, \ldots, X^{n+1} \in \mathbb{R}$ be real-valued random variables with sample mean $\bar{X}^{n+1}$ and sample standard deviation $\bar{S}^{n+1} > 0$. Define $Z^i = (X^i - \bar{X}^{n+1})/\bar{S}^{n+1}$ for $i \in [n+1]$. Then,*

$$|Z^i| \le \frac{n}{\sqrt{n+1}} \quad \textit{almost surely}, \quad \forall i \in [n+1].$$

This inequality is strict if $X^1, \ldots, X^{n+1}$ are almost-surely distinct.

Proof Fix any $i \in [n+1]$. Let $\bar{X}^{-i} = \frac{1}{n}\sum_{k \ne i} X^k$ be the sample mean of the random variables excluding the i-th one, and

$$\bar{S}^{-i} = \sqrt{\frac{1}{n}\sum_{k \ne i}(X^k - \bar{X}^{-i})^2}.$$

Then, it follows from Lemma S1 that

$$|Z^i| = \frac{|X^i - \bar{X}^{n+1}|}{\bar{S}^{n+1}} = \frac{\left|X^i - \frac{1}{n+1}\sum_{k=1}^{n+1} X^k\right|}{\sqrt{\frac{1}{n}\sum_{k=1}^{n+1}(X^k - \bar{X}^{n+1})^2}} = \frac{\frac{n}{n+1}|X^i - \bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i - \bar{X}^{-i})^2 + (\bar{S}^{-i})^2}}.$$

But $\bar{S}^{-i} \ge 0$, so we must have

$$|Z^i| \le \frac{\frac{n}{n+1}|X^i - \bar{X}^{-i}|}{\sqrt{\frac{1}{n+1}(X^i - \bar{X}^{-i})^2}} = \frac{n}{\sqrt{n+1}}.$$

If $X^1, \ldots, X^{n+1}$ are almost-surely distinct, we know $\bar{S}^{-i} > 0$ almost-surely for any $i \in [n+1]$ thus this inequality becomes strict. ■

B.2 Proofs of Main Results

Proof [of Theorem 1] This is a well-known result, whose proof is included for completeness. By definition of $\hat{C}(\mathbf{X}^{n+1})$ in (5),

$$\mathbf{Y}^{n+1} \notin \hat{C}(\mathbf{X}^{n+1}) \Longleftrightarrow S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n).$$

Moreover, $S^1, \ldots, S^{n+1}$ are exchangeable because $\mathbf{E}^1, \ldots, \mathbf{E}^{n+1}$ are exchangeable and Φ is fixed. Therefore, the proof is complete by applying the fundamental "quantile inflation lemma" of conformal prediction (e.g., see Angelopoulos et al. (2024)), which states

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \leq \alpha.$$

Moreover, it is also a standard result that, if $S^1, \ldots, S^{n+1}$ are almost-surely distinct,

$$\mathbb{P}\left[S^{n+1} > \hat{Q}_{1-\alpha}(S^1, \ldots, S^n)\right] \geq \alpha - \frac{1}{n+1}.$$

■

Proof [of Lemma 2] From Assumption 1, we know that $E_j^1, \ldots, E_j^{n+1}$ are almost-surely distinct for every $j \in [d]$ and $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost-surely for all $j \in [d]$, which also ensures S_{oracle}^{n+1} is well-defined. To prove this lemma, fix any $c \in \mathbb{R}$ and note that

$$\begin{aligned}
&S_{\text{oracle}}^{n+1} \leq c \\
&\iff E_j^{n+1} \leq c\,\hat{\sigma}_j^{\text{oracle}} + \hat{\mu}_j^{\text{oracle}}, \quad \forall j \in [d] \\
&\iff E_j^{n+1} - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i,\right)^2} - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i \leq 0, \quad \forall j \in [d] \\
&\iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n}\sum_{i=1}^{n+1}\left(E_j^i - \frac{1}{n+1}\sum_{i=1}^{n+1} E_j^i,\right)^2} \leq 0, \quad \forall j \in [d] \\
&\iff \underbrace{\frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) - c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2}}_{=: g_j(\mathbf{E}^{n+1})} \leq 0, \quad \forall j \in [d]
\end{aligned}$$

where the last implication follows from Lemma S1. Everything except $\mathbf{E}^{n+1}$ here is observed, so we can solve this inequality for $\mathbf{E}^{n+1}$ coordinate-wise.

Fix any $j \in [d]$. It follows from the above that

$$g_j(\mathbf{E}^{n+1}) \leq 0 \iff \frac{n}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right) \leq c\sqrt{\frac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2}. \tag{S1}$$

The solution of this inequality in E_j^{n+1} depends on the range of c. There are two cases:

1. if $c \geq 0$, then (S1) holds if and only if *either* $0 \leq E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *or* $E_j^{n+1} \geq \hat{\mu}_j^{\text{obs}}$ and

$$\left(\frac{n}{n+1}\right)^2 \left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \leq c^2 \left(\frac{1}{n+1}(E_j^{n+1} - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

2. if $c < 0$, then (S1) holds if and only if $E_j^{n+1} < \hat{\mu}_j^{\text{obs}}$ *and*

$$\left(\frac{n}{n+1}\right)^2 \left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 \geq c^2 \left(\tfrac{1}{n+1}\left(E_j^{n+1} - \hat{\mu}_j^{\text{obs}}\right)^2 + (\hat{\sigma}_j^{\text{obs}})^2\right).$$

The solution in both cases relies on solving a quadratic equality in the form $h(x) = 0$, where:

$$h(x) := \left(\frac{n}{n+1}\right)^2 \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - c^2 \tfrac{1}{n+1} \left(x - \hat{\mu}_j^{\text{obs}}\right)^2 - c^2 (\hat{\sigma}_j^{\text{obs}})^2 = 0.$$

This equality can be reorganized into a more recognized quadratic form

$$h(x) = Ax^2 + Bx + K = 0,$$

where

$$\begin{aligned}
A &= \left(\frac{n}{n+1}\right)^2 - \frac{c^2}{n+1},\\
B &= 2\hat{\mu}_j^{\text{obs}} \left(\frac{n}{n+1}\right)^2 - 2\hat{\mu}_j^{\text{obs}} \frac{c^2}{n+1},\\
K &= \left(\frac{n}{n+1}\right)^2 (\hat{\mu}_j^{\text{obs}})^2 - \frac{c^2}{n+1} (\hat{\mu}_j^{\text{obs}})^2 - c^2 (\hat{\sigma}_j^{\text{obs}})^2.
\end{aligned}$$

The solution set is:

$$h(x) = 0 \iff x = \begin{cases} \hat{\mu}_j^{\text{obs}} \pm \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c^2 < \frac{n^2}{n+1},\\ \hat{\mu}_j^{\text{obs}}, & c^2 = \frac{n^2}{n+1} \text{ and } \hat{\sigma}_j^{\text{obs}} = 0,\\ \text{no real solution}, & \text{otherwise.} \end{cases}$$

We first note the event $\hat{\sigma}_j^{\text{obs}} = 0$ occurs if and only if $E_j^1 = \ldots = E_j^n$, which has probability zero by Assumption 1. For the rest, we note that $c^2 < \frac{n^2}{n+1}$ implies $A > 0$ thus $h(x) = Ax^2 + Bx + K$ concave up for all $x \geq 0$. Therefore by analyzing the positiveness and negativeness of $h(x)$ relative to the solution set in our two cases, we know (S1) holds almost-surely if and only if

$$\begin{cases} \hat{\mu}_j^{\text{obs}} \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c \geq 0,\ c^2 < \frac{n^2}{n+1},\\ E_j^{n+1} < \infty, & c \geq 0,\ c^2 \geq \frac{n^2}{n+1},\\ 0 \leq E_j^{n+1} < \hat{\mu}_j^{\text{obs}}, & c \geq 0, \\ 0 \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & c < 0,\ c^2 < \frac{n^2}{n+1},\\ \text{no solution}, & c < 0,\ c^2 \geq \frac{n^2}{n+1}. \end{cases} \quad \text{almost-surely}$$

$$\iff \begin{cases} 0 \leq E_j^{n+1} \leq \hat{\mu}_j^{\text{obs}} + \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}, & 0 \leq c < \frac{n}{\sqrt{n+1}},\\ 0 \leq E_j^{n+1} < \infty, & c \geq \frac{n}{\sqrt{n+1}},\\ 0 \leq E_j^{n+1} \leq \max\left\{0, \hat{\mu}_j^{\text{obs}} - \hat{\sigma}_j^{\text{obs}} |c| \sqrt{\frac{(n+1)^2}{n^2 - (n+1)c^2}}\right\}, & -\frac{n}{\sqrt{n+1}} < c < 0,\\ \text{no solution}, & c \leq -\frac{n}{\sqrt{n+1}}. \end{cases} \quad \text{almost surely}$$

Interpreting the "no-solution" case as $E_j^{n+1} = 0$, we obtain precisely the expression for $\omega_j(c)$ given in the statement of the lemma. This notation adjustment is harmless because both $E_j^{n+1} = 0$ and $S_{\text{oracle}}^{n+1} \leq -n/\sqrt{n+1}$ are probability null sets by our Assumption 1 and Lemma S3, respectively. The monotonic nondecreasing property of this $\omega_j(c)$ follows immediately from the construction. ■

Proof [of Lemma 3] From Assumption 1, $E_j^1, \ldots, E_j^{n+1}$ are almost-surely distinct for every $j \in [d]$ and $0 < \hat{\sigma}_j^{\text{oracle}} < \infty$ almost-surely for all $j \in [d]$. This ensures S_{oracle}^i is well-defined for all $i \in [n+1]$. Moreover, we know from Lemma S3 that

$$\left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \frac{n}{\sqrt{n+1}},$$

almost-surely for all $i \in [n+1]$. Aggregating this inequality for all $j \in [d]$, we obtain

$$|S_{\text{oracle}}^i| = \left| \max_{1 \leq j \leq d} \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| \leq \max_{1 \leq j \leq d} \left| \frac{E_j^i - \hat{\mu}_j^{\text{oracle}}}{\hat{\sigma}_j^{\text{oracle}}} \right| < \max_{1 \leq j \leq d} \frac{n}{\sqrt{n+1}} = \frac{n}{\sqrt{n+1}},$$

almost-surely for all $i \in [n+1]$. Recall that the quantile $\hat{Q}_{1-\alpha}^{\text{oracle}}$ is defined as

$$\hat{Q}_{1-\alpha}^{\text{oracle}} := \lceil (1-\alpha)(n+1) \rceil \text{th smallest value in } \{S_{\text{oracle}}^1, \ldots, S_{\text{oracle}}^n, +\infty\},$$

so $\hat{Q}_{1-\alpha}^{\text{oracle}} < n/\sqrt{n+1}$ almost-surely if $n \geq 1/\alpha - 1$ and $\hat{Q}_{1-\alpha}^{\text{oracle}} = +\infty$ otherwise. ■

Proof [of Lemma 4] Fix any $j \in [d]$ and let

$$g(t_j, z_j) := \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)}.$$

Finding the supremum of this g over $z_j \geq 0$ is a simple univariate optimization problem. Expanding g in full, we get

$$g(t_j, z_j) = \frac{t_j - \hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}},$$

where the last equality follows from Lemma S1. Differentiating g with respect to z_j yields

$$g_{z_j}(t_j, z_j) = \frac{-\frac{1}{n+1}\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2} - \left(t_j - \frac{1}{n+1}(z_j + n\hat{\mu}_j^{\text{obs}})\right)\frac{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}}}{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\text{obs}})^2 + (\hat{\sigma}_j^{\text{obs}})^2}.$$

Solving $g_{z_j}(t_j, z_j^*) = 0$ with respect to z_j^* gives:

$$(\hat{\sigma}_j^{\text{obs}})^2 + (t_j - \hat{\mu}_j^{\text{obs}})(z_j^* - \hat{\mu}_j^{\text{obs}}) = 0 \quad \Longrightarrow \quad z_j^* = \hat{\mu}_j^{\text{obs}} - \frac{(\hat{\sigma}_j^{\text{obs}})^2}{t_j - \hat{\mu}_j^{\text{obs}}}.$$

There are three cases here for any fixed $t_j \geq 0$:

- Case 1: if $g(t_j, z_j^*)$ is indeed the global maximum and $z_j^* > 0$, then we can just take

$$\sup_{z_j \geq 0} g(t_j, z_j) = g(t_j, z_j^*).$$

- Case 2: if $g(t_j, z_j^*)$ is not the global maximum or $z_j^* < 0$, then the supremum must occur at either $z_j = 0$ or $z_j \to \infty$ due to the fact that g_{z_j} has only one critical point. Hence,

$$\sup_{z_j \geq 0} g(t_j, z_j) = \max\left\{g(t_j, 0), \lim_{z_j \to \infty} g(t_j, z_j)\right\} = \max\left\{g(t_j, 0), -\frac{1}{\sqrt{n+1}}\right\}.$$

- Case 3: if $t_j = \hat{\mu}_j^{\text{obs}}$, then the same derivation shows that the numerator of g_{z_j} reduces to $-\frac{1}{n+1}(\hat{\sigma}_j^{\text{obs}})^2 < 0$. This implies that g is monotonically decreasing and thus we take

$$\sup_{z_j \geq 0} g(\hat{\mu}_j^{\text{obs}}, z_j) = g(\hat{\mu}_j^{\text{obs}}, 0).$$

Collecting all candidates, we obtain the closed form:

$$\sup_{z_j \geq 0} g(t_j, z_j) = \max\left\{g\left(t_j, 0\right), g\left(t_j, z_j^* \mathbb{I}\left\{z_j^* \geq 0\right\}\right), -\frac{1}{\sqrt{n+1}}\right\}.$$

■

Proof [of Proposition 6] Throughout this proof we write $\tilde{C}(\mathbf{X}^{n+1}) := \hat{C}(\mathbf{X}^{n+1}; \hat{Q}_{1-\alpha}^{\text{oracle}})$ for the estimated oracle prediction set defined in (11). Let $\mathcal{P}$ be any (random) partition of $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1})$, so that $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) = \bigsqcup_{\mathcal{R} \in \mathcal{P}} \mathcal{R}$ almost-surely. Because $\hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \supseteq \tilde{C}(\mathbf{X}^{n+1})$ almost-surely, we know

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\iff \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \hat{C}^{\text{gwc}}(\mathbf{X}^{n+1}) \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}), \mathbf{Y}^{n+1} \in \mathcal{R} \\
&\iff \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R}.
\end{aligned}$$

If $\hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})$ satisfies the containment on the right-hand side of (19),

$$\mathcal{R} \supseteq \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \supseteq \tilde{C}(\mathbf{X}^{n+1}) \cap \mathcal{R},$$

then

$$\begin{aligned}
\mathbf{Y}^{n+1} \in \tilde{C}(\mathbf{X}^{n+1}) &\implies \exists \mathcal{R} \in \mathcal{P} : \mathbf{Y}^{n+1} \in \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\iff \mathbf{Y}^{n+1} \in \bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \\
&\implies \mathbf{Y}^{n+1} \in \text{Rect}_{\mathcal{P}}\left(\bigsqcup_{\mathcal{R} \in \mathcal{P}} \hat{C}^{\text{lwc}}(\mathbf{X}^{n+1}; \mathcal{R})\right) = \hat{C}_{\mathcal{P}}^{\text{lwc}}(\mathbf{X}^{n+1}),
\end{aligned}$$

where the last implication holds because $\mathrm{Rect}_{\mathcal{P}}(A) \supseteq A$ by definition. Since we already know that $\mathbb{P}[\mathbf{Y}^{n+1} \in \widetilde{C}(\mathbf{X}^{n+1})] \geq 1-\alpha$, by Theorem 1 and the dual relation (12), this proves that

$$\mathbb{P}\left[\mathbf{Y}^{n+1} \in \hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1})\right] \geq 1-\alpha.$$

Finally, the containment $\hat{C}^{\mathrm{lwc}}_{\mathcal{P}}(\mathbf{X}^{n+1}) \subseteq \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1})$ almost-surely is immediate from the definition of $\mathrm{Rect}_{\mathcal{P}}$ in (20), because

$$\hat{C}^{\mathrm{lwc}}(\mathbf{X}^{n+1}; \mathcal{R}) \subseteq \mathcal{R} \subseteq \hat{C}^{\mathrm{gwc}}(\mathbf{X}^{n+1}),$$

for every $\mathcal{R} \in \mathcal{P}$. ∎

Proof [of Lemma 7] Fix any $h \in [n+1]^d$. Suppose $\mathcal{Y}^h \neq \emptyset$, then $\mathcal{E}^h \neq \emptyset$ and, by Equation (21),

$$\mathcal{E}^h = \bigtimes_{j=1}^{d} \mathcal{E}^h_j, \quad \text{where} \quad \mathcal{E}^h_j := \left[E_j^{(h_j-1)}, U_j^{h_j}\right) \neq \emptyset, \quad \forall j \in [d].$$

Fix any $\mathbf{t} \in \mathbb{R}^d_+$. Similar to the proof of Lemma 4, we have to solve a univariate optimization problem. We first notice that for any $j \in [d]$,

$$\sup_{\mathbf{z} \in \mathcal{E}^h} \frac{t_j}{\hat{\sigma}_j(z_j)} = \sup_{z_j \in \mathcal{E}^h_j} \frac{t_j}{\hat{\sigma}_j(z_j)} = \frac{t_j}{\inf_{z_j \in \mathcal{E}^h_j} \hat{\sigma}_j(z_j)}.$$

By Lemma S1, we know that

$$\hat{\sigma}_j(z_j) = \sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2},$$

which is continuous in z_j, so the infimum above is attained on the closure $\overline{\mathcal{E}^h_j} = \left[E_j^{(h_j-1)}, U_j^{h_j}\right]$; clearly $\hat{\sigma}_j(z_j)$ is minimized there at

$$z_j^* = \underset{z_j \in \overline{\mathcal{E}^h_j}}{\operatorname{argmin}} |z_j - \hat{\mu}_j^{\mathrm{obs}}| = \begin{cases} \hat{\mu}_j^{\mathrm{obs}}, & \text{if } \hat{\mu}_j^{\mathrm{obs}} \in \overline{\mathcal{E}^h_j}, \\ E_j^{(h_j-1)}, & \text{if } \hat{\mu}_j^{\mathrm{obs}} < E_j^{(h_j-1)}, \\ U_j^{h_j}, & \text{if } \hat{\mu}_j^{\mathrm{obs}} > U_j^{h_j}. \end{cases}$$

Hence, $r_j(h_j) = \hat{\sigma}_j(z_j^*)$ is exactly the infimum proposed. For the second part, denote

$$g(z_j) := \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} = \frac{\frac{1}{n+1}(z_j + n\hat{\mu}_j^{\mathrm{obs}})}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2}} = \frac{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}}) + \hat{\mu}_j^{\mathrm{obs}}}{\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2}},$$

then we want to find

$$\inf_{\mathbf{z} \in \mathcal{E}^{\mathrm{gwc}}} g(z_j) = \inf_{z_j \in [0, \omega_j(\hat{Q}^{\mathrm{gwc}}_{1-\alpha})]} g(z_j).$$

Differentiating $g(z_j)$ in z_j, we get

$$g'(z_j) = \frac{\frac{1}{n+1}(\hat{\sigma}_j^{\mathrm{obs}})^2 - \frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})\hat{\mu}_j^{\mathrm{obs}}}{(\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2)\sqrt{\frac{1}{n+1}(z_j - \hat{\mu}_j^{\mathrm{obs}})^2 + (\hat{\sigma}_j^{\mathrm{obs}})^2}}.$$

Setting this to zero gives the critical point

$$z_j^* = \hat{\mu}_j^{\text{obs}} + \frac{(\hat{\sigma}_j^{\text{obs}})^2}{\hat{\mu}_j^{\text{obs}}} > 0.$$

Let $\epsilon > 0$, then we see that

$$\begin{aligned}
\text{sign}(g'(z_j^* + \epsilon)) &= \text{sign}\left(-\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = -1, \\
\text{sign}(g'(z_j^* - \epsilon)) &= \text{sign}\left(\tfrac{1}{n+1}\epsilon\hat{\mu}_j^{\text{obs}}\right) = +1.
\end{aligned}$$

This implies that $g(z_j^*)$ is always a local maximum. Hence, if $z_j^* \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]$, then

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = \min\left\{ g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} g(z_j) \right\};$$

if $z_j^* > \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})$, then g is nondecreasing on $[0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]$ and

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = g(0).$$

But in either case, we only need to compare the boundaries in order to find the infimum. Therefore, it follows that

$$\inf_{z_j \in [0, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})]} g(z_j) = \min\left\{ g(0), \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} g(z_j) \right\}.$$

This is true for every $j \in [d]$, so we must have

$$\begin{aligned}
\hat{\Phi}_h^{\text{lwc}}(\mathbf{t}) &= \max_{1 \le j \le d} \left\{ \sup_{\mathbf{z} \in \mathcal{E}^h} \frac{t_j}{\hat{\sigma}_j(z_j)} - \inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} g(z_j) \right\} \\
&= \max_{1 \le j \le d} \left\{ \frac{t_j}{r_j(h_j)} - \min\left\{ \frac{\hat{\mu}_j(0)}{\hat{\sigma}_j(0)}, \lim_{z_j \uparrow \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})} \frac{\hat{\mu}_j(z_j)}{\hat{\sigma}_j(z_j)} \right\} \right\}.
\end{aligned}$$

■

Proof [of Lemma 8] Fix any $j \in [d]$ and suppose $\mathcal{Y}^{h^*} \neq \emptyset$. There are two parts to this lemma, which we prove separately. Throughout, for any $k \in [n+1]$ we let $h^*(j,k)$ denote the unique multi-index in $\text{Row}_j(h^*)$ whose j-th coordinate equals k; that is, $h_j^*(j,k) = k$ and $h_l^*(j,k) = h_l^*$ for all $l \neq j$.

Part I. We aim to prove

$$\sup_{h \in [n+1]^d} B_j^h = \sup_{h \in \text{Row}_j(h^*)} B_j^h = B_j^{h^{[j]}}. \tag{S2}$$

We first notice that the first supremum can be decomposed into suprema over surfaces:

$$\sup_{h \in [n+1]^d} B_j^h = \sup_{k \in [n+1]} \sup_{h \in \text{Surface}_j(k)} B_j^h,$$

where

$$\text{Surface}_j(k) := \left\{ h \in [n+1]^d : h_j = k \right\}.$$

Fix any $k \in [n+1]$ and note that $h^*(j,k) \in \text{Surface}_j(k) \cap \text{Row}_j(h^*)$. Since this holds for every $k \in [n+1]$, to prove (S2) the main challenge is to show that:

$$\sup_{h \in \text{Surface}_j(k)} B_j^h = B_j^{h^*(j,k)}. \tag{S3}$$

To prove (S3), recall from the definition in (22) and Lemma 7 that

$$\hat{\Phi}_h^{\text{lwc}}(\mathbf{t}) = \max_{1 \le l \le d} \left\{ \frac{t_l}{r_l(h_l)} - c_l \right\},$$

where

$$r_l(h_l) = \min \left\{ \hat{\sigma}_l(z_l) : z_l \in \overline{\mathcal{E}_l^h} \right\}, \quad \overline{\mathcal{E}_l^h} := \left[E_l^{(h_l-1)}, U_l^{h_l} \right], \qquad c_l = \inf_{\mathbf{z} \in \mathcal{E}^{\text{gwc}}} \frac{\hat{\mu}_l(z_l)}{\hat{\sigma}_l(z_l)}.$$

Without loss of generality, assume $\mathcal{Y}^h \neq \emptyset$ for all $h \in \text{Surface}_j(k)$. This does not affect the final result because $\mathcal{Y}^h = \emptyset$ leads to $B_j^h = 0$, which can be safely excluded from the supremum calculation. Then notice that, for all $h \in \text{Surface}_j(k)$, we have $r_j(h_j) = r_j(k)$ and $r_l(h_l) \ge \hat{\sigma}_l^{\text{obs}} = r_l(h_l^*)$ for all $l \neq j$, the latter because $\hat{\mu}_l^{\text{obs}} \in \overline{\mathcal{E}_l^{h^*}}$ by the definition of the mean index h^*. Therefore, for any fixed $\mathbf{t}$,

$$\hat{\Phi}_h^{\text{lwc}}(\mathbf{t}) = \max_{1 \le l \le d} \left\{ \frac{t_l}{r_l(h_l)} - c_l \right\} \le \max \left(\max_{l \neq j} \left\{ \frac{t_l}{\hat{\sigma}_l^{\text{obs}}} - c_l \right\}, \frac{t_j}{r_j(k)} - c_j \right) = \hat{\Phi}_{h^*(j,k)}^{\text{lwc}}(\mathbf{t}).$$

Because this holds point-wise for all $\mathbf{t}$, we can extend the inequality to the corresponding conformal quantile and conclude that, almost surely for all $h \in \text{Surface}_j(k)$,

$$\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h) \le \hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})$$

and therefore, by the monotonicity of the function ω_j from Lemma 2,

$$\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h)) \le \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})).$$

Therefore, it follows that, for all $h \in \text{Surface}_j(k)$,

$$\begin{aligned}
B_j^h &= \begin{cases} \min \left\{ U_j^k, \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h)) \right\}, & \text{if } U_j^k,\ \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^h)) > E_j^{(k-1)}, \\ 0, & \text{otherwise,} \end{cases} \\
&\le \begin{cases} \min\left\{ U_j^k, \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})) \right\}, & \text{if } U_j^k,\ \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k)})) > E_j^{(k-1)}, \\ 0, & \text{otherwise,} \end{cases} \\
&= B_j^{h^*(j,k)}.
\end{aligned}$$

This completes the proof of (S3).

The second equality in (S2) is trivial. We can easily see that for any $k \in [n]$, if $B_j^{h^*(j,k+1)} \neq 0$, then

$$U_j^{k+1},\ \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,k+1)})) \geq E_j^{(k)} \geq \min\left\{E_j^{(k)}, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})\right\} = U_j^k.$$

Hence, it follows that $B_j^{h^*(j,k+1)} \geq B_j^{h^*(j,k)}$ regardless of whether $B_j^{h^*(j,k)} = 0$ or not. Then finding the supremum among all $k \in [n+1]$ simplifies to finding the right-most nonzero $B_j^{h^*(j,k)}$, which is exactly how $B_j^{h^{[j]}}$ is defined.

Part II. In this part, we aim to show that if $B_j^{h^*(j,m_j)} = 0$ for some $m_j \geq h_j^*$, then

$$B_j^{h^*(j,N)} = 0, \quad \forall N \geq m_j.$$

There are two cases that could lead to $B_j^{h^*(j,m_j)} = 0$:

- If $U_j^{m_j} = \min\left\{E_j^{(m_j)}, \omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}})\right\} \leq E_j^{(m_j-1)}$, then $\omega_j(\hat{Q}_{1-\alpha}^{\text{gwc}}) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)}$ for any $N \geq m_j$ by the nature of order statistics. Hence, $U_j^N \leq E_j^{(N-1)}$ for all $N \geq m_j$, which further implies $B_j^{h^*(j,N)} = 0$ for all $N \geq m_j$.

- If $\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)})) \leq E_j^{(m_j-1)}$, then we claim that

$$\omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,N)})) \leq \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)})) \leq E_j^{(m_j-1)} \leq E_j^{(N-1)},$$

  which then implies $B_j^{h^*(j,N)} = 0$ for all $N \geq m_j$. This is actually trivial because $m_j \geq h_j^*$ implies that both $\overline{\mathcal{E}_j^{h^*(j,m_j)}}$ and $\overline{\mathcal{E}_j^{h^*(j,N)}}$ lie to the right of $\hat{\mu}_j^{\text{obs}}$, with the latter no closer to it than the former, so that

$$r_j(m_j) = \min_{z_j \in \mathcal{E}_j^{h^*(j,m_j)}} \hat{\sigma}_j(z_j) \leq \min_{z_j \in \mathcal{E}_j^{h^*(j,N)}} \hat{\sigma}_j(z_j) = r_j(N),$$

  by Lemma S2 and the definition of $\mathcal{E}_j^h$. This then leads to

$$\begin{aligned}
&\hat{\Phi}_{h^*(j,m_j)}^{\text{lwc}}(\mathbf{t}) \geq \hat{\Phi}_{h^*(j,N)}^{\text{lwc}}(\mathbf{t}), \quad \forall \mathbf{t} \\
&\implies \hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,N)}) \leq \hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)}) \\
&\implies \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,N)})) \leq \omega_j(\hat{Q}_{1-\alpha}^{\text{lwc}}(\mathcal{Y}^{h^*(j,m_j)})).
\end{aligned}$$

■

Appendix C. Additional Numerical Results

This section presents the results of additional numerical experiments, including comparisons with a broader range of alternative approaches. These experiments are conducted using the same protocol and performance metrics described in Section 4 unless specified otherwise.

C.1 Additional Results from the Experiments of Section 4

Here we present additional results from the numerical experiments described in Section 4, corresponding to six different noise distributions. These results are qualitatively consistent with those summarized in Section 4: our method performs consistently well, producing relatively tight prediction sets with valid coverage.

C.1.1 Gaussian Noise (heterogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gaussian)	Emp. Copula	0.714 (0.079)	1.68e+10(1.16e+10)
	Point CHR	0.933 (0.066)	4.04e+12(1.38e+13)
	TSCP (Ours)	0.910 (0.053)	**1.83e+11(2.74e+11)**
	Unscaled Max	0.908 (0.047)	2.22e+13(4.77e+13)
(50, 10, Gaussian)	Emp. Copula	0.816 (0.057)	3.64e+10(2.22e+10)
	Point CHR	0.919 (0.057)	4.85e+11(1.72e+12)
	TSCP (Ours)	0.904 (0.047)	**9.42e+10(8.82e+10)**
	Unscaled Max	0.897 (0.043)	1.16e+13(1.10e+13)
(100, 10, Gaussian)	Emp. Copula	0.819 (0.039)	2.58e+10(9.90e+09)
	Point CHR	0.901 (0.041)	9.74e+10(9.22e+10)
	TSCP (Ours)	0.903 (0.034)	**6.59e+10(3.43e+10)**
	Unscaled Max	0.908 (0.027)	1.09e+13(6.93e+12)
(300, 10, Gaussian)	Emp. Copula	0.872 (0.019)	3.73e+10(8.11e+09)
	Point CHR	0.899 (0.025)	5.60e+10(2.05e+10)
	TSCP (Ours)	0.899 (0.018)	**4.93e+10(1.17e+10)**
	Unscaled Max	0.901 (0.020)	8.68e+12(3.29e+12)
(500, 10, Gaussian)	Emp. Copula	0.886 (0.018)	4.16e+10(8.27e+09)
	Point CHR	0.900 (0.022)	5.13e+10(1.49e+10)
	TSCP (Ours)	0.901 (0.016)	**4.81e+10(9.67e+09)**
	Unscaled Max	0.903 (0.016)	8.54e+12(2.46e+12)

Table S3: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, as in Figure 3. All results are averaged over 200 trials with their one standard deviation reported in parenthesis. The target coverage level is 0.9. Coverage below this target is highlighted in red. For volume, lower is better. Smallest volume size (among those with valid theoretical coverage) is highlighted in **bold**. TSCP yields the tightest volume while maintaining valid coverage.

C.1.2 Gaussian Noise (homogeneous)

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Unit Gaussian)	Emp. Copula	0.714 (0.079)	4.64e+03(3.20e+03)
	Point CHR	0.933 (0.066)	1.11e+06(3.80e+06)
	TSCP (Ours)	0.910 (0.053)	5.04e+04(7.56e+04)
	Unscaled Max	0.904 (0.054)	**2.19e+04(1.94e+04)**
(50, 10, Unit Gaussian)	Emp. Copula	0.816 (0.057)	1.00e+04(6.12e+03)
	Point CHR	0.919 (0.057)	1.34e+05(4.75e+05)
	TSCP (Ours)	0.904 (0.047)	2.60e+04(2.43e+04)
	Unscaled Max	0.901 (0.045)	**1.68e+04(1.15e+04)**
(100, 10, Unit Gaussian)	Emp. Copula	0.819 (0.039)	7.11e+03(2.73e+03)
	Point CHR	0.901 (0.041)	2.68e+04(2.54e+04)
	TSCP (Ours)	0.903 (0.034)	1.82e+04(9.46e+03)
	Unscaled Max	0.900 (0.035)	**1.50e+04(7.38e+03)**
(300, 10, Unit Gaussian)	Emp. Copula	0.872 (0.019)	1.03e+04(2.23e+03)
	Point CHR	0.899 (0.025)	1.54e+04(5.66e+03)
	TSCP (Ours)	0.899 (0.018)	1.36e+04(3.23e+03)
	Unscaled Max	0.898 (0.018)	**1.28e+04(3.07e+03)**
(500, 10, Unit Gaussian)	Emp. Copula	0.886 (0.018)	1.15e+04(2.28e+03)
	Point CHR	0.900 (0.022)	1.41e+04(4.11e+03)
	TSCP (Ours)	0.901 (0.016)	1.33e+04(2.67e+03)
	Unscaled Max	0.899 (0.017)	**1.26e+04(2.50e+03)**

Table S4: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, as in Figure 4. Other details are as in Table S3. In this case, Unscaled Max yields the tightest volume while maintaining valid coverage, while TSCP remains a competitive second.

C.1.3 Laplace Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Laplace)	Emp. Copula	0.712 (0.079)	3.82e+12(4.91e+12)
	Point CHR	0.938 (0.060)	5.57e+16(2.71e+17)
	TSCP (Ours)	0.908 (0.060)	**3.14e+14(9.93e+14)**
	Unscaled Max	0.897 (0.053)	6.36e+15(1.99e+16)
(50, 10, Laplace)	Emp. Copula	0.819 (0.056)	1.36e+13(1.75e+13)
	Point CHR	0.917 (0.050)	6.64e+14(1.81e+15)
	TSCP (Ours)	0.906 (0.045)	**7.32e+13(1.60e+14)**
	Unscaled Max	0.901 (0.041)	3.34e+15(5.12e+15)
(100, 10, Laplace)	Emp. Copula	0.811 (0.039)	5.96e+12(3.94e+12)
	Point CHR	0.896 (0.039)	4.74e+13(6.07e+13)
	TSCP (Ours)	0.898 (0.030)	**2.64e+13(1.94e+13)**
	Unscaled Max	0.895 (0.029)	1.77e+15(1.78e+15)
(300, 10, Laplace)	Emp. Copula	0.871 (0.020)	1.11e+13(4.03e+12)
	Point CHR	0.895 (0.023)	2.01e+13(1.12e+13)
	TSCP (Ours)	0.896 (0.017)	**1.74e+13(6.40e+12)**
	Unscaled Max	0.896 (0.017)	1.38e+15(7.08e+14)
(500, 10, Laplace)	Emp. Copula	0.882 (0.016)	1.29e+13(4.10e+12)
	Point CHR	0.896 (0.021)	1.83e+13(8.95e+12)
	TSCP (Ours)	0.896 (0.016)	**1.64e+13(5.68e+12)**
	Unscaled Max	0.897 (0.017)	1.36e+15(5.51e+14)

Table S5: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Laplace* distribution; i.e., $\epsilon_j \sim \text{Laplace}(10 - j + 1)$ for all $j \in [10]$. Other details are as in Table S3.

C.1.4 Mixture Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Mixed)	Emp. Copula	0.715 (0.075)	2.69e+11(3.04e+11)
	Point CHR	0.935 (0.062)	2.56e+15(1.98e+16)
	TSCP (Ours)	0.913 (0.051)	**9.53e+12(3.12e+13)**
	Unscaled Max	0.903 (0.052)	3.46e+13(6.74e+13)
(50, 10, Mixed)	Emp. Copula	0.821 (0.058)	7.07e+11(5.96e+11)
	Point CHR	0.925 (0.048)	6.10e+13(3.06e+14)
	TSCP (Ours)	0.906 (0.044)	**2.57e+12(3.91e+12)**
	Unscaled Max	0.901 (0.044)	1.98e+13(1.68e+13)
(100, 10, Mixed)	Emp. Copula	0.819 (0.035)	3.82e+11(1.98e+11)
	Point CHR	0.898 (0.042)	2.76e+12(4.47e+12)
	TSCP (Ours)	0.900 (0.029)	**1.23e+12(7.54e+11)**
	Unscaled Max	0.905 (0.028)	1.65e+13(9.89e+12)
(300, 10, Mixed)	Emp. Copula	0.874 (0.022)	6.47e+11(2.25e+11)
	Point CHR	0.902 (0.025)	1.27e+12(7.23e+11)
	TSCP (Ours)	0.900 (0.019)	**9.05e+11(3.48e+11)**
	Unscaled Max	0.900 (0.020)	1.36e+13(5.22e+12)
(500, 10, Mixed)	Emp. Copula	0.886 (0.016)	7.26e+11(1.82e+11)
	Point CHR	0.897 (0.020)	1.01e+12(5.42e+11)
	TSCP (Ours)	0.898 (0.014)	**8.17e+11(2.05e+11)**
	Unscaled Max	0.902 (0.016)	1.35e+13(3.73e+12)

Table S6: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Mixture* distribution; i.e., $\epsilon_j \sim \frac{1}{2}\text{Laplace}(10 - j + 1) + \frac{1}{2}\mathcal{N}(0, (10 - j + 1)^2)$ for all $j \in [10]$. Other details are as in Table S3.

C.1.5 Gamma Noise

(n, d, Noise)	Method	Coverage	Volume
(30, 10, Gamma)	Emp. Copula	0.714 (0.067)	5.33e-02(6.85e-02)
	Point CHR	0.929 (0.058)	7.55e+02(4.32e+03)
	TSCP (Ours)	0.904 (0.056)	**3.77e+00(1.59e+01)**
	Unscaled Max	0.898 (0.051)	2.69e+15(6.18e+15)
(50, 10, Gamma)	Emp. Copula	0.826 (0.048)	2.20e-01(2.78e-01)
	Point CHR	0.927 (0.049)	5.01e+01(3.36e+02)
	TSCP (Ours)	0.910 (0.041)	**1.15e+00(2.56e+00)**
	Unscaled Max	0.902 (0.039)	1.94e+15(3.54e+15)
(100, 10, Gamma)	Emp. Copula	0.818 (0.038)	1.04e-01(1.01e-01)
	Point CHR	0.906 (0.038)	1.28e+00(2.38e+00)
	TSCP (Ours)	0.900 (0.030)	**4.65e-01(5.82e-01)**
	Unscaled Max	0.897 (0.030)	9.67e+14(7.86e+14)
(300, 10, Gamma)	Emp. Copula	0.873 (0.019)	1.75e-01(1.15e-01)
	Point CHR	0.895 (0.024)	3.74e-01(3.70e-01)
	TSCP (Ours)	0.898 (0.018)	**2.67e-01(1.69e-01)**
	Unscaled Max	0.895 (0.018)	7.04e+14(3.45e+14)
(500, 10, Gamma)	Emp. Copula	0.885 (0.015)	1.89e-01(1.06e-01)
	Point CHR	0.896 (0.020)	3.20e-01(2.33e-01)
	TSCP (Ours)	0.897 (0.016)	**2.36e-01(1.35e-01)**
	Unscaled Max	0.893 (0.015)	6.37e+14(2.53e+14)

Table S7: Performance of the proposed TSCP method and alternative conformal prediction approaches on simulated data with $d = 10$ outcome variables, and noise following a *Gamma* distribution; i.e., $\epsilon_j \sim \text{Gamma}(1, 10 - j + 1)$ for all $j \in [10]$. Other details are as in Table S3.

C.2 Comparison of Alternative Implementations of TSCP

The additional methods considered in this section include (i) *Naive*, the heuristic plug-in rescaling method that uses location and scale parameters estimated directly from the calibration data without further adjustments, summarized in Algorithm S3; (ii) *TSCP-S*, the inefficient data-splitting variant summarized in Algorithm S4; (iii) *TSCP-GWC*, our conservative global-worst-case method summarized in Algorithm S5; (iv) *TSCP-LWC*, the aggregated union described in Section 3.4 but not enclosed by a rectangle.

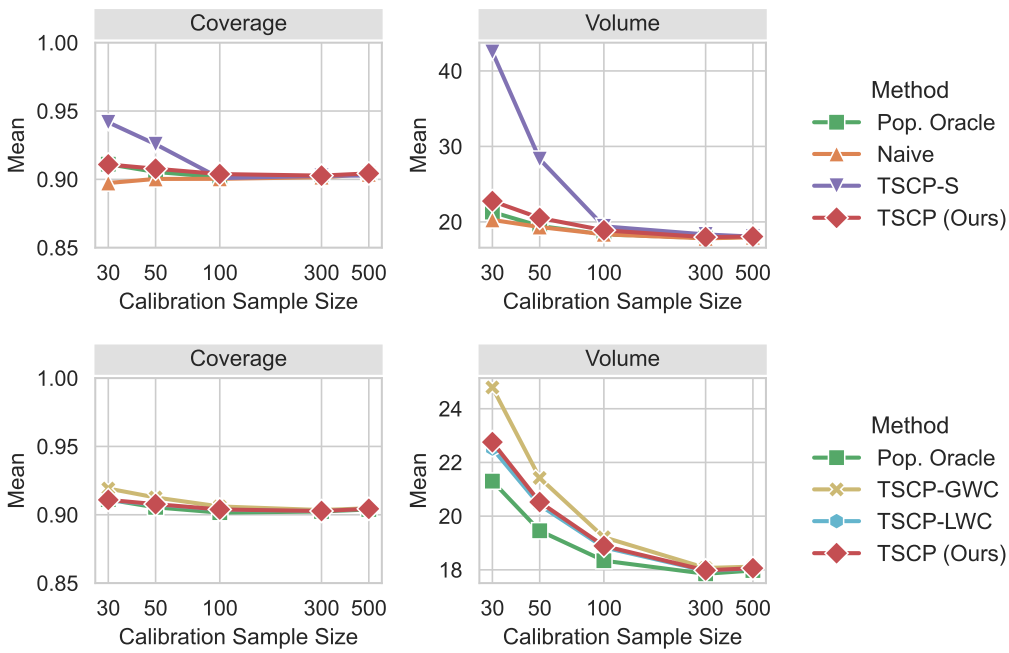


Figure S1: Performance of TSCP and alternative approaches on simulated data with $d = 2$ targets and heterogeneous Laplace noise. The target coverage level is 90%. Results are averaged over 200 trials. All methods achieve near-nominal coverage and approximate *Pop. Oracle* as the calibration sample size increases. TSCP tracks TSCP-LWC closely, with an almost indistinguishable average volume, highlighting the tightness of its rectangular enclosure.

All methods compared in Figure S1 tend to approximate *Pop. Oracle* well as n increases, with TSCP and TSCP-LWC being the two most stable and fastest approximations among all methods with theoretical coverage. These results also show that TSCP encloses the aggregated union TSCP-LWC very tightly, and that TSCP-GWC is conservative, though it still performs better than the data-splitting variant TSCP-S. Despite lacking a finite-sample coverage guarantee, the *Naive* approach performs well, as also shown in Figure S2.

Figure S3 better highlights the prohibitive computational cost of TSCP-LWC, by reporting on the results of similar experiments where a small sample size is fixed, $|\mathcal{D}_{\text{cal}}| = 30$, while the outcome dimensions are gradually increased: $d \in \{2, 3, 4, 5, 10, 20, 30\}$.

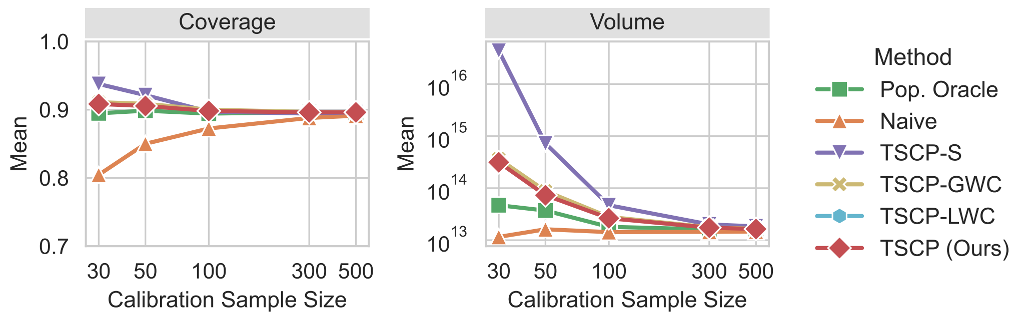


Figure S2: Performance of TSCP and alternative approaches on simulated data with $d = 10$ targets. Other details are as in Figure S1. TSCP-LWC is omitted from this figure due to its prohibitive cost with even moderately high-dimensional outcomes.

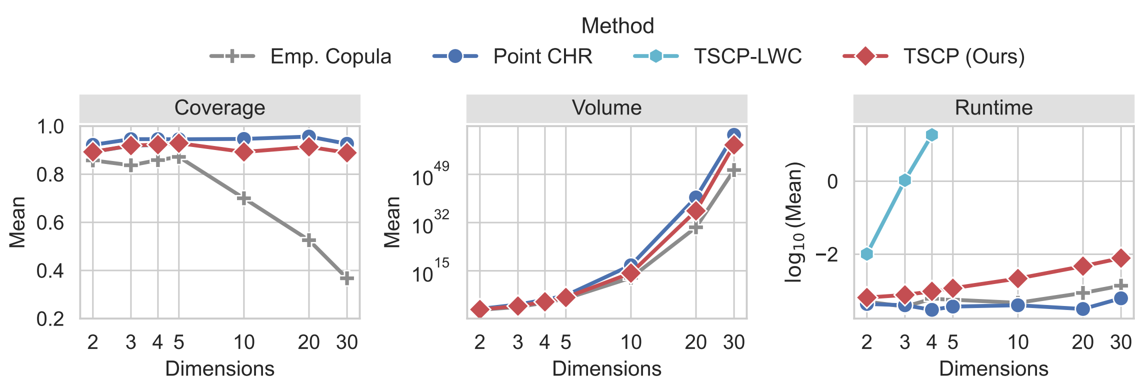
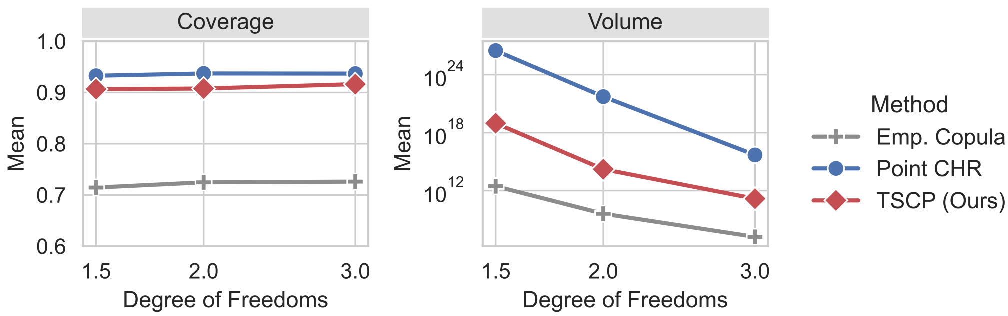


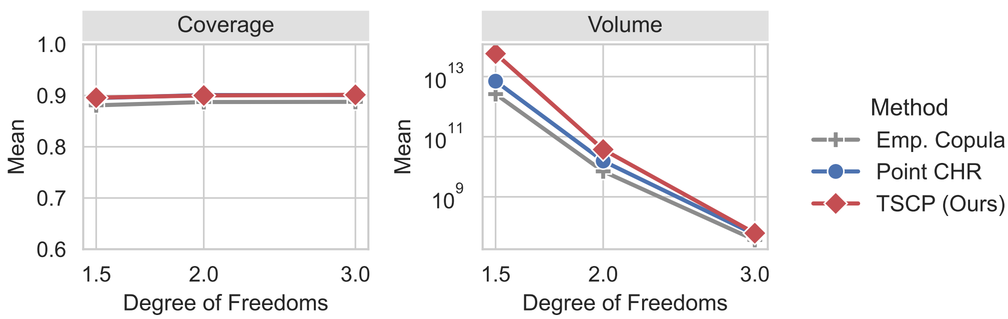
Figure S3: Performance of TSCP and alternative approaches on simulated data with fixed sample size $|\mathcal{D}_{\text{cal}}| = 30$, as a function of the number of output dimensions d. Other details are as in Figure S1. These results are averaged over 30 independent repetitions. The runtime results are presented in seconds on a $\log_{10}$ scale. The runtime of all methods except TSCP-LWC scales well with number of dimensions. TSCP-LWC is only applied when $d \in \{2, 3, 4\}$ due to the exponential growth of its runtime.

C.3 Robustness to Heavy-Tailed Data

We test the performance of TSCP on simulated heavy-tailed data with $d = 10$ outcomes and noise following a Student's-t distribution; i.e., $Y_j \mid \mathbf{X} \sim f_j(\mathbf{X}) + \epsilon_j$ with $\epsilon_j \sim t_{\mathrm{DF}}$ and $\mathrm{DF} \in \{1.5, 2, 3\}$ degrees of freedom. For $1 < \mathrm{DF} \leq 2$, the Student's-t distribution leads to residuals with infinite population variance, violating Assumption 1. Nonetheless, TSCP continues to achieve the desired coverage level even in this extreme setting, although with some loss of informativeness.



(a) Small sample, $|\mathcal{D}_{\mathrm{cal}}| = 30$. Output dimensions $d = 10$.



(b) Large sample, $|\mathcal{D}_{\mathrm{cal}}| = 500$. Output dimensions $d = 10$.

Figure S4: Performance of the proposed TSCP method and other approaches on simulated data with noise following a Student's-t distribution with varying degrees of freedom, using small (a) and large (b) sample sizes. The target coverage level is 0.9. All methods except *Emp. Copula* have near-nominal coverage in all settings. TSCP maintains valid marginal coverage even on very heavy-tailed data. In the small sample setting, Point CHR suffers more in terms of prediction set size due to its inefficient use of calibration data, although it can outperform TSCP in large samples.